

Preregistration: Megastudy testing 20 information interventions to strengthen trust in climate scientists in the US

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Trust in climate is strongly associated with belief in climate change and support for policies to mitigate climate change. Strengthening trust in climate scientists may therefore be an important lever for accelerating climate action. However, there is currently little evidence on how to strengthen trust in climate scientists. We aim to address this gap by systematically identifying, testing, and comparing information based interventions. To this end, we are conducting a megastudy—a large experiment in which many different interventions are tested simultaneously within a large sample, compared to the same control group, and using the same outcome measures. The study will be conducted in the United States, where public beliefs about climate change are highly polarized, and where climate research is particularly threatened by funding cuts. We will test 20 interventions, ranging from reading about climate science to discussing the topic with LLMs. The interventions were selected from 107 intervention proposals submitted by researchers in response to an open call to collaborate. No data has been collected at the moment of registration.

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i Note

If you are reading this as a pdf, a more reader-friendly html version is available on the [project website](#)

No data has been collected at the moment of registration.

Introduction

Climate change is considered one of the most pressing societal issues (Rogelj et al. 2023), necessitating urgent action at both the policymaking and individual level (Calvin et al. 2023). To believe in human made climate change, and in the fact that climate change is a problem, requires trusting climate scientists: As we cannot observe for ourselves how human actions affect the climate, we have to rely on climate scientists data, models, and interpretations.

Correlational evidence stresses the crucial role of trust for beliefs and attitudes about climate change. People with higher trust in scientists in general tend to be more likely to accept the scientific consensus on global warming (Bogert et al. 2024), to have more accurate beliefs about climate change (Ejaz, Vu, and Fletcher 2025), and to support climate policies (Cologna and Siegrist 2020; Hornsey et al. 2016). A large-scale study across 55 countries found that trust in climate scientists was the strongest predictor of belief in climate change and support for climate policies (Todorova et al. 2025). Strengthening trust in climate scientists may therefore be an important lever to accelerate climate action.

But how to strengthen trust in climate scientists? Trust relationships are complex and slow to build—in general, humans are not easily swayed into trusting others (Mercier 2020). Trust in science and scientists is no exception: structural issues, such as a systematic underrepresentation of certain groups among scientists (Druckman et al. 2025) are likely to be a major cause of distrust in science and these issues don’t have easy-to-implement fixes. At the same time, science communication plays a crucial role in building trust (Intemann 2023), and behavioral sciences offer tools to improve communication strategies. Several large-scale investigations have drawn on insights from the behavioral sciences to design messages aimed at changing people’s belief in, concern over, and intentions to act against climate change (Voelkel et al. 2026; Vlasceanu et al. 2024; Goldwert et al. 2026; Sinclair et al. 2025; Huber et al. 2026).

However, there is currently little evidence on which science communication strategies can strengthen trust in climate scientists. One study showed that when receiving a message about the need for changes in individual behavior or in public policy to address climate change, people tend to trust climate scientists less (Palm, Bolsen, and Kingsland 2020). Another study showed that participants perceived climate scientists as more skilled—a dimension of trustworthiness—after reading a text about the longstanding history and foundations of climate science (Orchinik et al. 2024). Exercising intellectual humility—acknowledging the limits of one’s knowledge—has been shown to increase trust in a fictive virologist and a fictive climate scientists (Koetke et al. 2024).

More indirect evidence comes from studies investigating how to strengthen trust in scientists in general. For scientists in general, studies have found that using open-science practices (Rosman et al. 2022; Song, Markowitz, and Taylor 2022), highlighting successful replications (Hendriks, Kienhues, and Bromme 2020), and communicating uncertainty (Schneider et al. 2022) can increase trust. One study found that presenting participants with an infographic about the scientific process slightly increased trust in scientists (Agle et al. 2021). In another study,

participants tended to ascribe more expertise—a dimensions of trustworthiness—to scientists, when they present two-sided rather than one-sided arguments (Hendriks, Janssen, and Jucks 2023). One study found that reading short biographies of scientists made participants perceive the scientists as more trustworthy, presumably demonstrating benevolence (Hautea, Besley, and Choung 2024). Not all intervention studies have produced positive results. For example, a registered report testing messages tailored to conservatives in the US did not change their trust in scientists (Gligorić, Van Kleef, and Rutjens 2025).

There are two issues regarding these findings on strengthening trust in scientists in general: First, it is not clear how these findings transfer to climate scientists specifically, as people tend to perceive climate scientists differently. While trust in scientists is moderately high globally (Cologna et al. 2025), climate scientists have consistently been found to be less trusted than scientists in general (Ghasemi et al. 2025; Schröder 2023; Schug, Bilandzic, and Kinnebrock 2024) and scientists from other disciplines (Druckman et al. 2024; Schröder 2023; Schug, Bilandzic, and Kinnebrock 2024; Pfänder and Mercier 2025; Gligorić, Kleef, and Rutjens 2024). Second, current evidence on how to strengthen trust in scientists suffers from a lack of comparability. It is difficult to compare the effectiveness of different communication strategies on trust in scientists, because of differences in sample selection, outcome measures, and experimental design (Pfänder, Mede, and Cologna 2026). One example is uncertainty communication: While communicating scientific uncertainties appears to have a positive impact on trust in science on average, the results are highly context-dependent, with some studies finding negative effects (Schuster and Scheu 2026; Bles et al. 2020). It is hard to make sense of these mixed results and pinpoint contextual causes given the very different research designs of the studies.

Here, we aim to address these issues by systematically identifying, testing, and comparing different communication strategies to strengthen trust in climate scientists. To do so, we run a megastudy—“a massive field experiment in which many different treatments are tested synchronously in one large sample using a common, objectively measured outcome” (Milkman et al. 2021). The megastudy comprises 20 text-based interventions, selected from 107 submissions to an open call for collaboration, reviewed and edited by 13 members of an expert advisory board. While the main outcome of interest of this megastudy is trust in climate scientists, we will also assess the interventions’ impact on various secondary (e.g., donations to a scientific association, support for public funding of climate research) and tertiary outcome variables (e.g., belief in climate change, support for climate policies). We will also investigate heterogeneous treatment effects (e.g., whether certain interventions are more effective among Democrats or Republicans).

The results of our study have the potential to inform the communication strategies of a wide range of actors—e.g., universities, research institutes, governments, or NGOs—looking to rally support in the fight against climate change.

Research Questions

Our primary research question is: Which information interventions significantly increase trust in climate scientists? We hypothesize that our interventions will significantly increase trust in climate scientists compared to the control group.

There are three groups of secondary research questions: First, how do the information interventions affect other outcomes of interests? Second, are the effects of the information interventions moderated by other variables? Third, do the information intervention effects persist, such that they can still be detected a week later?

Ethics

The study was approved by the Institutional Review Board at Stanford University in the United States (Protocol ID: IRB-85756) and at ETH (Protocol ID: 26 ETHICS-093) in Switzerland. All participants will provide informed consent and will be paid for their participation. Participants will be randomly assigned to their experimental condition and will be blind to the study design. We will not use deception. Because the study will be conducted online, there will be no interaction between the experimenter and participants.

Interventions

We used an open call for interventions designed to strengthen trust in climate scientists. We received 105 intervention proposals¹ from 80 different research teams² (25 teams submitted two interventions), involving 73 different researchers from 65 different institutions.

An expert reviewer team selected 20 promising treatments for experimental testing. The expert reviewer team consisted of three members of the research team and an advisory board of 9 researchers and one practitioner. The selection process was fully anonymized—only the research lead could link submissions to their authors, and did not participate in the rating process. For details on the selection process, see the supplemental material.

Table 1 provides an overview of the interventions. In the control condition, participants will be randomly assigned to read one of three neutral texts unrelated to climate change. These texts cover (a) the history of neckties, (b) the rules of baseball, and (c) different types of dances. The detailed stimuli for all intervention and control conditions can be found in the questionnaire, a separate document attached to this pre-registration.

¹We originally received 107 proposals, but two got retracted from the authoring teams shortly after the submission deadline.

²some “teams” are in fact individual researchers—24 of the 105 submissions are single-authored

Table 1: Overview of interventions included in the megastudy.

	Intervention Title	Summary
Collaboration and peer-review		
1	Interview Prof. Maraun	Climate scientist Prof. Douglas Maraun at the University of Graz in Austria stresses the collaborative and self-correcting process of climate science.
2	Peer-review	What makes climate science trustworthy is the process of independent peer-review.
Scientific methods and results		
3	Measurement & modeling (1)	Climate scientists use sophisticated measurement and computational modeling techniques to surveil climate and predict how it changes.
4	Measurement & modeling (2)	Climate scientists are primarily natural scientists (e.g., biologists, physicists). They use sophisticated tools and quantitative methods to measure and predict climate change.
5	Model accuracy	This is an edited version of a real news article showcasing that even old climate models, despite some flaws, were remarkably correct in predicting global warming.
Applications and impact		
6	Portrait Prof. Cherry	Todd Cherry, a scientist focused on climate issues, is integrated in his local community, and does work that is relevant for this community.
7	Extreme weather predictions	Showcases how climate science predicts and helps adapting to different extreme weather events (blizzards, floods, wildfires). Takes into account a participant's state and addresses the extreme weather event most common in that state.
Others' endorsement		
8	Corporate reliance	Insurance companies and large corporations rely on climate scientists' projections.
9	Former skeptics	Former climate change skeptics Jennifer Rukavina (television meteorologist) and Bob Inglis (former Republican congressman) explain how they came to change their mind.
Values		
10	Value similarity	The quiz 'Which Type of Climate Scientist Are You?' highlights dimensions of climate scientists' trustworthiness. By providing participants with their personalized climate scientists profile, the intervention intends to create perceptions of value similarity and identification.

Continued on next page

Table 1: Overview of interventions included in the megastudy. (Continued)

	Intervention Title	Summary
11	Interview Prof. Sebille	Prof. Erik van Sebille, a climate scientist and oceanographer at Utrecht University, Netherlands, mentions harmful consequences of climate change on oceans and humans, and how he cares about preventing these consequences.
	LLM-chatbot	
12	LLM chatbot (1)	LLM-chatbot
13	LLM chatbot (2)	LLM-chatbot
14	LLM-chatbot (3)	LLM-chatbot
	Other	
15	Social justice	In the United States, the wealthiest 10% of the population are responsible for roughly 40% of the country’s total greenhouse gas emissions. Climate scientists provide evidence to hold the emitters accountable.
16	Funding	Correcting potential misperceptions on the amount and sources of climate science funding. Showcases that climate science receives relatively little public and private funding.
17	Oil industry misinformation	Oil companies have spent decades financing large propaganda campaigns to cast doubt on the existence climate change and the credibility of climate scientists.
18	High public trust	Correct potential misperceptions of how many Americans trusts climate scientists. A majority of Americans trusts climate scientists at least to some extent.
19	Scientist community helpers	Climate scientists are members of local communities and their work helps local communities in times of climate disasters (e.g., floods and wildfires).
20	Consensus	Correcting potential misperceptions on the level of agreement among climate scientists on climate change, and climate change related information.

Procedure and Design

Experiment

Participants first provide informed consent before taking part in the study. They are then informed that they need to qualify for the study. During the qualification phase, participants begin by completing a demographic questionnaire, which includes measures of gender, age,

race, education, income, household size, social class, and residential area. Throughout the survey, participants receive a prompt if they leave a question unanswered, giving them a second opportunity to provide a response. In general, no items are strictly forced, with the following exception: questions on gender, age, and race are mandatory, as these variables are used to implement quotas to ensure that the sample is broadly representative of the U.S. population. The qualification phase also includes an initial attention check embedded within the demographic questionnaire. Participants who fail this attention check are immediately excluded from the study. Participants then proceed to answer questions about their partisan identity and their religion. They are then presented with a second attention check. Failure to pass this check also results in immediate exclusion from the remainder of the study.

Participants who provide the required demographic information and pass both attention checks are informed that they qualify for the study. They are then told that the study concerns opinions about climate change and climate scientists and are provided with a definition of climate scientists (“Climate scientists study changes in the Earth’s climate over time and how they might affect the planet in the future.”).

Next, participants complete measures of pre-treatment variables, including belief in climate change, trust in climate scientists (single-item measure), perceived alienation from climate science, and need for epistemic autonomy.

Participants are then randomly assigned to one of 21 experimental conditions (20 intervention conditions and one control condition). In each of the intervention conditions, participants receive a short text-based informational intervention about climate science or climate scientists. In the control condition, participants will be randomly assigned to read one of three neutral texts unrelated to climate change. These texts cover (a) the history of neckties, (b) the rules of baseball, and (c) different types of dances. Using multiple control texts reduces the risk that unintended characteristics of any single text influence the outcome variables.

After being exposed to the content of their respective conditions, participants will complete a set of outcome measures. There are three groups of outcomes. The primary outcome is a multidimensional measure of trust in climate scientists. The second block of outcomes assesses attitudes directly related to climate science and scientists, views on the role of climate scientists in policymaking, trust in public climate research institutions, and trust and distrust in climate scientists (single-item measures). In addition, the block of secondary outcomes contains two behavioral measures: For one, participants are asked to allocate money between themselves and the American Meteorological Society (AMS). For the other, participants are given the opportunity to subscribe to the free version of a climate scientist’s newsletter (Katharine Hayhoe’s “Talking Climate” newsletter). The third block of outcomes consists of variables related more generally to climate change and climate change mitigation, including belief in and concern about climate change, support for climate change mitigation policies, and individual mitigation behaviors.

Follow-up survey

Approximately one week after the main experiment, a follow-up survey will be conducted to assess the persistence of the most effective interventions. Participants from the five intervention conditions that show the strongest effects, as well as participants from the control group, will be invited to complete the outcome measures again. Note that the follow-up survey will not include the newsletter signup outcome again. The reason is that people have either signed up or not previously, and we won't measure whether they have signed out again after a week.

Measures

Note

Throughout this manuscript, to reduce computation time, we only run analyses on a couple of outcomes for illustration. This is why some outcomes do not appear in the tables or plots.

We measure trust in climate scientists using a multidimensional scale as our primary outcome. To ensure that all interventions would focus on this target, we did not communicate secondary or tertiary outcomes to research teams during the call for intervention submissions (see below). Secondary and tertiary outcomes capture downstream attitudinal and behavioral consequences of trust, as well as related constructs. The distinction between secondary and tertiary outcomes reflects theoretical proximity to the primary outcome — secondary outcomes such as institutional trust and funding perceptions are more directly related to trust in climate scientists, while tertiary outcomes such as behavioral intentions and policy support are more distal. Practical relevance also plays a role: behavioral measures and funding support are designated secondary due to their direct policy implications. We distinguish between outcome groups primarily to contextualize findings in light of multiple comparison concerns. Although we statistically correct for multiple comparisons within each outcome separately, the sheer number of outcomes increases the risk of false positives across the study. Being transparent about our ordering of theoretical interest is intended to help readers interpret findings appropriately: effects on tertiary outcomes should be treated with greater caution than effects on primary or secondary outcomes. Details on all measures are provided in the questionnaire, included as a separate document.

Primary outcome

The primary outcome is a multidimensional measure of trust in climate scientists, aggregating across four subdimensions: competence, integrity, benevolence, and openness. Each subdimension is measured with three items on a 0–100 slider scale and averaged into a subdimension

score; the four subdimension scores are then averaged into the composite. The subdimensions are also analyzed separately in the item-level analyses (see below).

Secondary outcomes

Secondary outcomes capture a single-item measure for trust and distrust in climate scientists, perceptions of climate science funding³, views on the appropriate policy role of scientists, and trust in scientific and governmental institutions. Trust and distrust are measured with single post-treatment items. Institutional trust is measured across five institutions (EPA, NASA, NOAA, universities, and federal government) and averaged into a composite, but we will report item-level analyses, too (see below).

The secondary outcomes also include two behavioral measures. Donation to the American Meteorological Society is a real monetary allocation on a 0–10 scale and is analyzed using OLS. Newsletter signup is a binary outcome (whether the participant signed up for a climate science newsletter) and is analyzed separately.

Tertiary outcomes

Tertiary outcomes capture broader climate-related attitudes and behavioral intentions. These include prior climate change belief, general concern about climate change (mean of three items), general climate policy support, specific climate policy support (mean of seven items), and individual-level climate mitigation behaviors (mean of six items). For concern, we additionally contrast absolute concern (mean of items 1–2) with relative concern (item 3: importance of climate change relative to other issues), as the relative item taps a conceptually distinct construct. Item-level analyses for specific policies and behaviors are reported separately.

Item-level analyses

For some of the included scales, we expect items to be heterogeneous, because they are conceptually different. For these scales—trust dimensions, institutional trust, specific climate policies, and individual behaviors—we will report the results on the aggregate measure (the mean of all items) in the manuscript, but also report item-level analyses in the supplemental materials.

The code chunk below defines key sets of variables used throughout the code in this pre-registration.

³Note that the variable science funding will need to be reverse-coded. In the questionnaire, higher values correspond to perceptions of currently “too much” funding. Reverse-coding will result in higher values indicating currently perceptions of currently “too little” funding, or, in other words, support for more funding.

```

outcomes <- c(
  # Primary
  "trust_multidimensional",
  # Secondary
  "trust_post",
  "distrust_post",
  "funding_perceptions",
  "policy_role_mean",
  "inst_trust_mean",
  "donation_ams",
  # Tertiary
  "belief_post",
  "concern_mean",
  "policy_general",
  "policy_specific_mean", # composite - items analyzed separately in appendix
  "behavior_mean"         # composite - items analyzed separately in appendix
) # note that this excludes behavioral outcomes
# (donation_ams, newsletter_signup)
# which are treated separately due to different scales
# and measurement approaches

# behavioral outcomes - secondary, but require separate treatment
# donation_ams: real money allocation (0-10 scale)
# newsletter_signup: binary - modeled with logistic regression throughout
behavioral_outcomes <- c(
  "donation_ams",
  "newsletter_signup"
)

# item-level outcomes - analyzed separately in appendix via linear mixed models
trust_dimensions <- c(
  "trust_competence",
  "trust_integrity",
  "trust_benevolence",
  "trust_openness"
)

inst_trust_items <- c(
  "inst_trust_epa",
  "inst_trust_nasa",
  "inst_trust_noaa",
  "inst_trust_universities",

```

```

    "inst_trust_federal_gov"
  )

policy_specific_items <- c(
  "policy_specific_1", "policy_specific_2", "policy_specific_3",
  "policy_specific_4", "policy_specific_5", "policy_specific_6",
  "policy_specific_7"
)

behavior_items <- c(
  "behavior_meat", "behavior_transport", "behavior_solar",
  "behavior_fly", "behavior_talk", "behavior_donate"
)

# to reduce computation time for this preregistration
outcomes_illustrative <- c("trust_multidimensional",
                           "donation_ams",
                           "funding_perceptions",
                           "policy_general")

secondary_outcomes <- c(
  "trust_post",
  "distrust_post",
  "funding_perceptions",
  "policy_role_mean",
  "inst_trust_mean"
)

tertiary_outcomes <- c(
  "belief_post",
  "concern_mean",
  "policy_general",
  "policy_specific_mean",
  "behavior_mean"
)

demographics <- c(
  "age", "gender", "race", "education",
  "income", "social_class", "urban_rural"
)

```

```

covariates <- c("age", "gender", "race")

moderators <- c(
  demographics,
  "party",
  "religion", "born_again", "religiosity",
  "belief_pre",
  "trust_pre"
)

# to reduce computation time for this preregistration
moderators_illustrative <- c("party", "gender", "social_class", "education",
                             "belief_pre", "age")

```

Sampling Plan

Participants will be recruited from a national, non-probability opt-in panel of US residents provided by CloudResearch. We will use cross quotas on gender $\times$ age and gender $\times$ race/ethnicity to approximate the US adult population. Quota targets (Table 2) are derived from the 2024 vintage of the US Census Bureau’s Population Estimates Program, accessed via the `tidycensus` R package. For each age group and racial/ethnic category, we set separate targets for male and female participants. Participants selecting “Other” as their gender are not subject to quotas since the Census Bureau does not provide population estimates for this category — their inclusion is therefore determined by natural panel availability. The table reports the total target percentage for each category, as well as the male and female breakdown within each category. Cross quotas constrain the joint distribution of gender with age and race/ethnicity, reducing the risk of imbalances such as too many young White women and too few older Black men that can arise with marginal quotas.

Table 2: Sampling quota targets derived from the 2024 US Census Bureau Population Estimates Program. Total (%) reflects the sum of the targeted size within each category of age and race. Male (%) and Female (%) show the gender breakdown within each category. Absolute numbers are based on a total sample size of $N = 22,000$

Category	Total	Male	Female
Age			
18-29	4435 (20.2%)	2259 (50.9%)	2176 (49.1%)
30-44	5730 (26.0%)	2891 (50.5%)	2839 (49.5%)
45-59	5038 (22.9%)	2503 (49.7%)	2535 (50.3%)
60+	6797 (30.9%)	3136 (46.1%)	3661 (53.9%)
Race / Ethnicity			
Asian / Asian American	1468 (6.7%)	694 (47.3%)	774 (52.7%)
Black / African American	2704 (12.3%)	1274 (47.1%)	1430 (52.9%)
Hispanic / Latino	3988 (18.1%)	2012 (50.4%)	1976 (49.6%)
Other	601 (2.7%)	293 (48.8%)	308 (51.2%)
White (non-Hispanic)	13240 (60.2%)	6517 (49.2%)	6723 (50.8%)

Data collection will be stopped as soon as we have collected complete responses from 22,000 participants (1,000 participants for each of the 20 treatment conditions, 2,000 participants for the control condition). Before treatment assignment, we will deploy a series of attention and bot detection checks. Participants will be informed right after the tests on whether they passed or not. Only those participants who have passed will be able to continue and complete the study.

No data has been collected at the moment of registration.

According to Monte Carlo power simulations, a sample size of $N = 22,000$ would allow us to detect our smallest effect of interest, Cohen’s $d = 0.15$, with statistical power of 94% (Figure 2). An effect size of Cohen’s $d = 0.15$ is typically considered small in behavioral sciences. In a pilot sample ($N = 76$), this minimal effect size of interest would translate to a 2.22 points change of the sample mean on a scale from 0, very low trust, to 100, very high trust (see Figure 1). This effect size falls into the range of effects found by other megastudies: For example, Voelkel et al. (2026) report successful interventions on climate related attitudes to range between 1 and 4 points on 100 point outcome scales. In the simulations, we adjusted p-values for multiple

testing via the Benjamini–Hochberg false discovery rate procedure—the same procedure we rely on for our analyses—and used $\alpha = .05$ as the cutoff point for statistical significance. More information on the power simulations can be found in the supplemental material.

This power analysis also provides a rough idea for the power of the follow-up survey: Assuming that the effect persists fully (i.e. the effect is the same as in the experiment), even with a relatively low retention rate of 60% (i.e. 600 participants per treatment arm) we would still be able to detect Cohen’s $d = 0.15$ with a power of 73%.

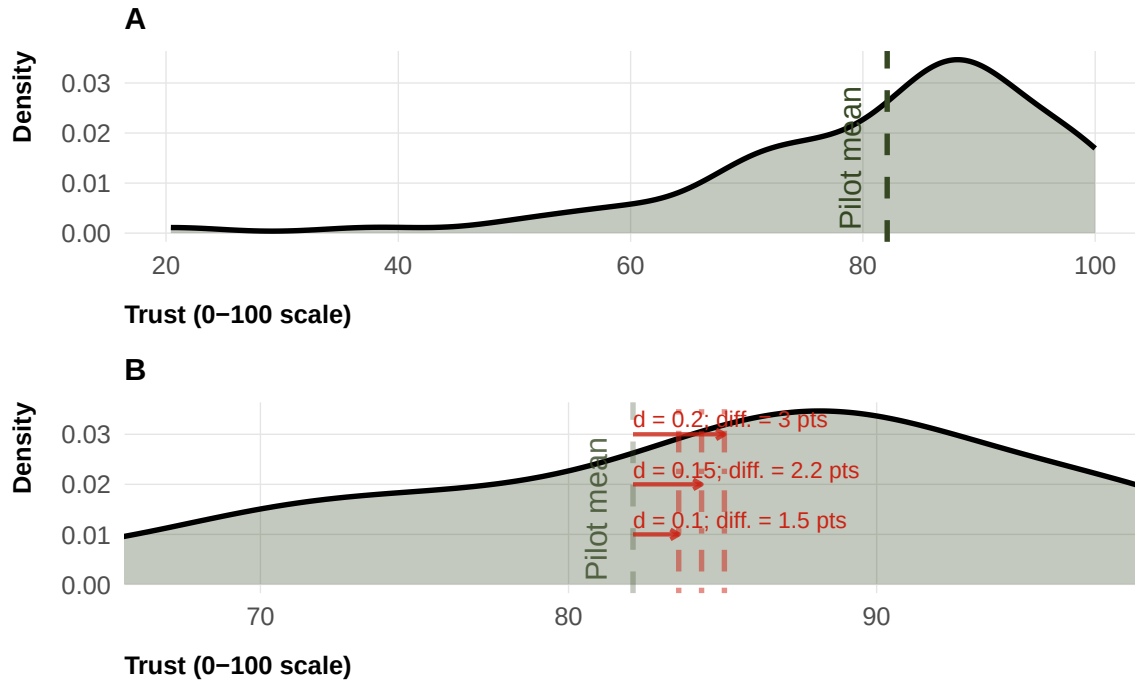


Figure 1: **A** Distribution of trust from the control condition (N=76) of a pilot study of one of the submitted interventions. **B** Translation of standardized effect sizes tested in the power simulation to the original 0-100 trust scale.

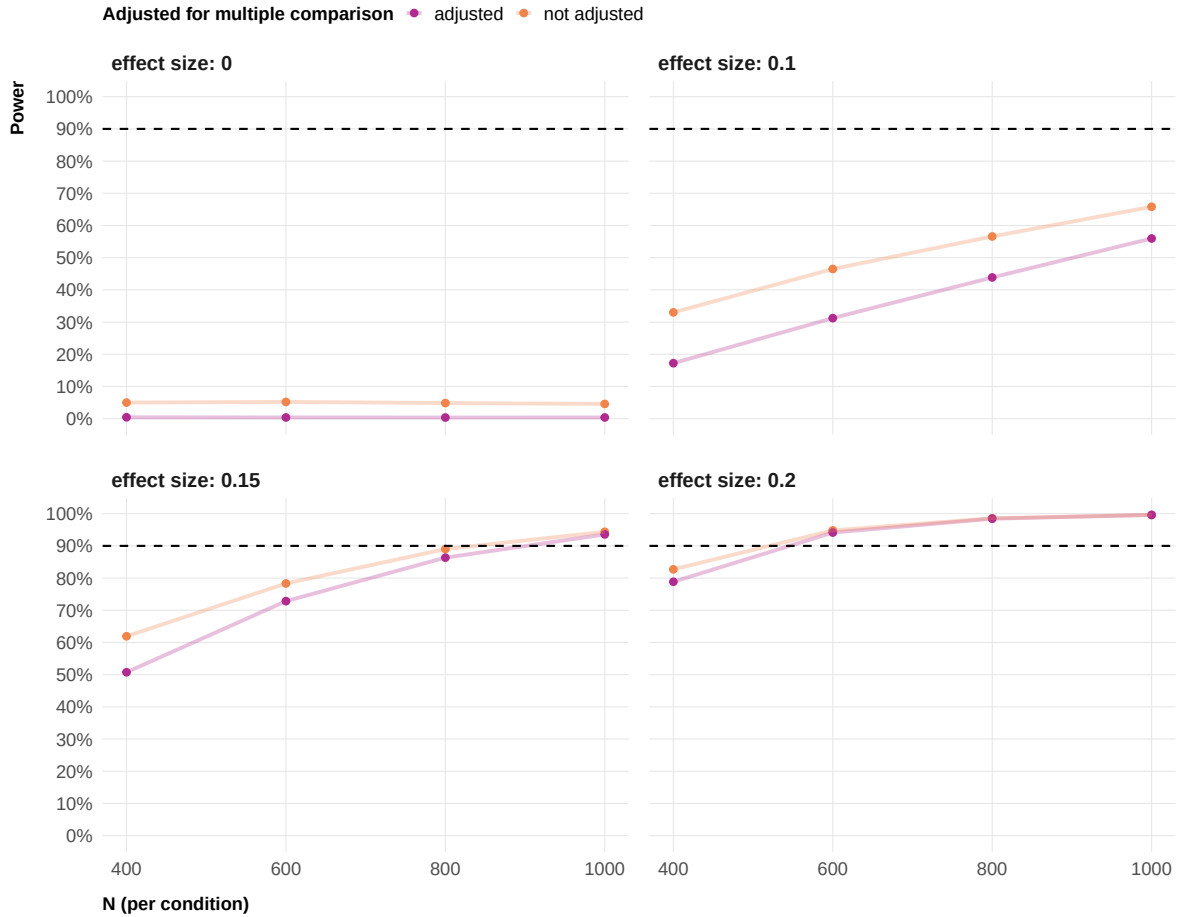


Figure 2: Results of the power simulation. Power is defined as the average share of statistically significant effects within a study, across 1,000 simulated studies. The plot shows power as a function of sample size (n per experimental condition, with equal sample size assumed for all 20 treatment conditions, and twice as many for the control condition). The two curves represent non-adjusted vs. adjusted p-values. The three facets represent different effect sizes.

Analysis plan

Exclusions

First, we do not allow for any individual participant taking part in our study several times. In cases with a duplicated participant ID, we will only keep the first case. Second, we will exclude participants who failed a series of attention and bot detection checks. These checks

will be run before treatment assignment to avoid post-treatment bias (Montgomery, Nyhan, and Torres 2018).

Treatment effects

We will test the effects of each of the treatments relative to the control condition with ordinary least squares regression. We will use heteroskedasticity-robust standard errors to ensure valid statistical inference in the presence of potentially unequal error variances across experimental conditions.

For all outcome variables, we will separately regress each post-treatment outcome on a categorical variable for experimental condition, using the control condition as the baseline category. The categorical condition variable will be represented as a series of dummy variables, one for each of the 20 interventions, with the control condition as the omitted reference category. To reduce residual variance and increase statistical power, we will include age, gender, and race as covariates in all models. These variables are used to implement sampling quotas and are therefore mandatory—they have no missing values and their inclusion carries no risk of reducing the analyzed sample size. We do not include other pre-treatment variables (e.g., single-item trust in climate change or belief in climate change) as covariates, even though they would likely explain additional outcome variance, because they are not mandatory and may have missing values. Including covariates with missing values causes listwise deletion⁴, which could significantly reduce the analyzed sample size.

For all continuous outcome variables, we estimate the following ordinary least squares (OLS) model.⁵

$$Y_i = \beta_0 + \sum_{k=1}^K \beta_k D_{ik} + \mathbf{X}_i \gamma + \varepsilon_i,$$

where Y_i is a continuous outcome, D_{ik} is a binary variable equal to 1 if participant i was assigned to intervention k , and 0 otherwise. The main control condition serves as the omitted reference category. $\mathbf{X}_i$ denotes the vector of covariates (gender, age, race), and ε_i is an error term. All models are estimated using ordinary least squares with heteroskedasticity-robust standard errors. All statistical tests are two-sided.

For each outcome, we test the null hypothesis $H_0 : \beta_k = 0$ for each intervention k , corresponding to no difference relative to the control condition. To account for multiple comparisons, we adjust p-values using the Benjamini–Hochberg (or false discovery rate, FDR) procedure across the 20 intervention-vs-control comparisons within each outcome separately. Although not all megastudies do this (e.g., Voelkel et al. 2024, 2026), researchers have stressed the importance of accounting for multiple comparison between the different treatment arms in megastudies

⁴With listwise deletion, only participants with valid values for *all* covariates are included in the model.

⁵The binary outcome of newsletter signup is handled separately and described below.

(Milkman et al. 2021; Milkman et al. 2022). Unlike some other megastudies (e.g., Goldwert et al. 2026) we will not apply additional corrections for multiple comparison across different outcomes, as we consider each outcome as an independent test.

```
run_main_treatment_model <- function(data,
                                     outcome,
                                     condition_var = "condition",
                                     covariates    = NULL,
                                     weights       = NULL,
                                     adjust_method = "BH") {

  # Formula
  rhs      <- paste(c(condition_var, covariates), collapse = " + ")
  model_formula <- as.formula(paste(outcome, "~", rhs))

  # Baseline (control) level
  baseline <- levels(data[[condition_var]])[1]

  # Fit
  fit <- lm(
    model_formula,
    data      = data,
    weights = if (!is.null(weights)) data[[weights]] else NULL
  )

  # Robust VCOV (HC2)
  vcov_robust <- sandwich::vcovHC(fit, type = "HC2")

  results <- lmtest::coeftest(fit, vcov = vcov_robust) |>
    broom::tidy(conf.int = TRUE) |>
    filter(str_detect(term, paste0("^", condition_var))) |>
    mutate(
      outcome      = outcome,
      condition    = str_remove(term, condition_var),
      baseline     = baseline,
      p.value_adjusted = p.adjust(p.value, method = adjust_method),
      significant_adjusted = case_when(
        p.value_adjusted < .001 ~ "****",
        p.value_adjusted < .01  ~ "***",
        p.value_adjusted < .05  ~ "**",
        TRUE                  ~ NA_character_
      )
    ) |>
```

```

    select(-term)

  return(results)
}

# check if the main control condition is correctly assigned to be the baseline
# levels(data$condition)

# calculate results for all outcome variables
main_model_results <- map_df(
  outcomes,
  ~ run_main_treatment_model(
    data = data,
    outcome = .x,
    condition_var = "condition",
    covariates = covariates
  )
)

```

Figure 3 shows a possible presentation of the estimated treatment effects for all outcome variables. We will present detailed model results in the appendix (see Table 5).

Binary outcome: newsletter signup

Newsletter signup is a binary outcome and is therefore modeled separately from the continuous outcomes. We estimate the effect of each intervention on the probability of signing up for a climate science newsletter using logistic regression. As with the continuous outcomes, we include gender, age, and race as covariates. Formally, our model is:

$$\log \left(\frac{P(Y_i = 1)}{1 - P(Y_i = 1)} \right) = \beta_0 + \sum_{k=1}^K \beta_k D_{ik} + \mathbf{X}_i \gamma + \varepsilon_i$$

where $P(Y_i = 1)$ is the probability of signing up for the newsletter. Coefficients are estimated on the log-odds scale. To facilitate interpretation, we report the average difference in predicted signup probability between each intervention and the control condition, computed on the probability scale using the `marginalEffects` package in R (Arel-Bundock, Greifer, and Heiss 2024), which handles the transformation of coefficients and standard errors from the log-odds scale to the probability scale. All estimates use heteroskedasticity-robust standard errors (HC2), and p-values are adjusted using the Benjamini–Hochberg procedure across the 20 intervention-vs-control comparisons.

```

run_main_treatment_model_binary <- function(data,
                                             outcome,
                                             condition_var = "condition",
                                             covariates      = NULL,
                                             weights         = NULL,
                                             adjust_method = "BH") {

  rhs      <- paste(c(condition_var, covariates), collapse = " + ")
  model_formula <- as.formula(paste(outcome, "~", rhs))
  baseline    <- levels(data[[condition_var]])[1]

  fit <- glm(
    model_formula,
    data      = data,
    family    = binomial(link = "logit"),
    weights   = if (!is.null(weights)) data[[weights]] else NULL
  )

  vcov_robust <- sandwich::vcovHC(fit, type = "HC2")

  # Log-odds results - for inference
  log_odds <- lmtest::coeftest(fit, vcov = vcov_robust) |>
    broom::tidy(conf.int = TRUE) |>
    filter(str_detect(term, paste0("^", condition_var))) |>
    mutate(
      outcome      = outcome,
      condition    = str_remove(term, condition_var),
      baseline     = baseline,
      p.value_adjusted = p.adjust(p.value, method = adjust_method),
      significant_adjusted = case_when(
        p.value_adjusted < .001 ~ "***",
        p.value_adjusted < .01  ~ "**",
        p.value_adjusted < .05  ~ "*",
        TRUE                  ~ NA_character_
      ),
      odds_ratio    = exp(estimate),
      or_conf.low   = exp(conf.low),
      or_conf.high  = exp(conf.high)
    ) |>
    select(-term)

  # Marginal effects - for interpretation (probability scale)

```

```

marginal_effects <- marginaleffects::avg_comparisons(
  fit,
  variables = condition_var,
  vcov      = vcov_robust,
  newdata = data |> select(all_of(c(outcome, condition_var, covariates)))
) |>
as_tibble() |>
select(contrast, estimate, conf.low, conf.high, p.value) |>
mutate(
  condition      = str_remove(contrast, " - .+$"),
  outcome        = outcome,
  baseline       = baseline,
  p.value_adjusted = p.adjust(p.value, method = adjust_method),
  significant_adjusted = case_when(
    p.value_adjusted < .001 ~ "***",
    p.value_adjusted < .01  ~ "**",
    p.value_adjusted < .05  ~ "*",
    TRUE                  ~ NA_character_
  )
)

list(
  log_odds      = log_odds,
  marginal_effects = marginal_effects
)
}

```

```

# calculate results for all outcome variables
main_model_newsletter_signup <- run_main_treatment_model_binary(
  data = data,
  outcome = "newsletter_signup",
  covariates = covariates
)

```

A possible visualization of the treatment effects on newsletter signup can be found in Figure 3.

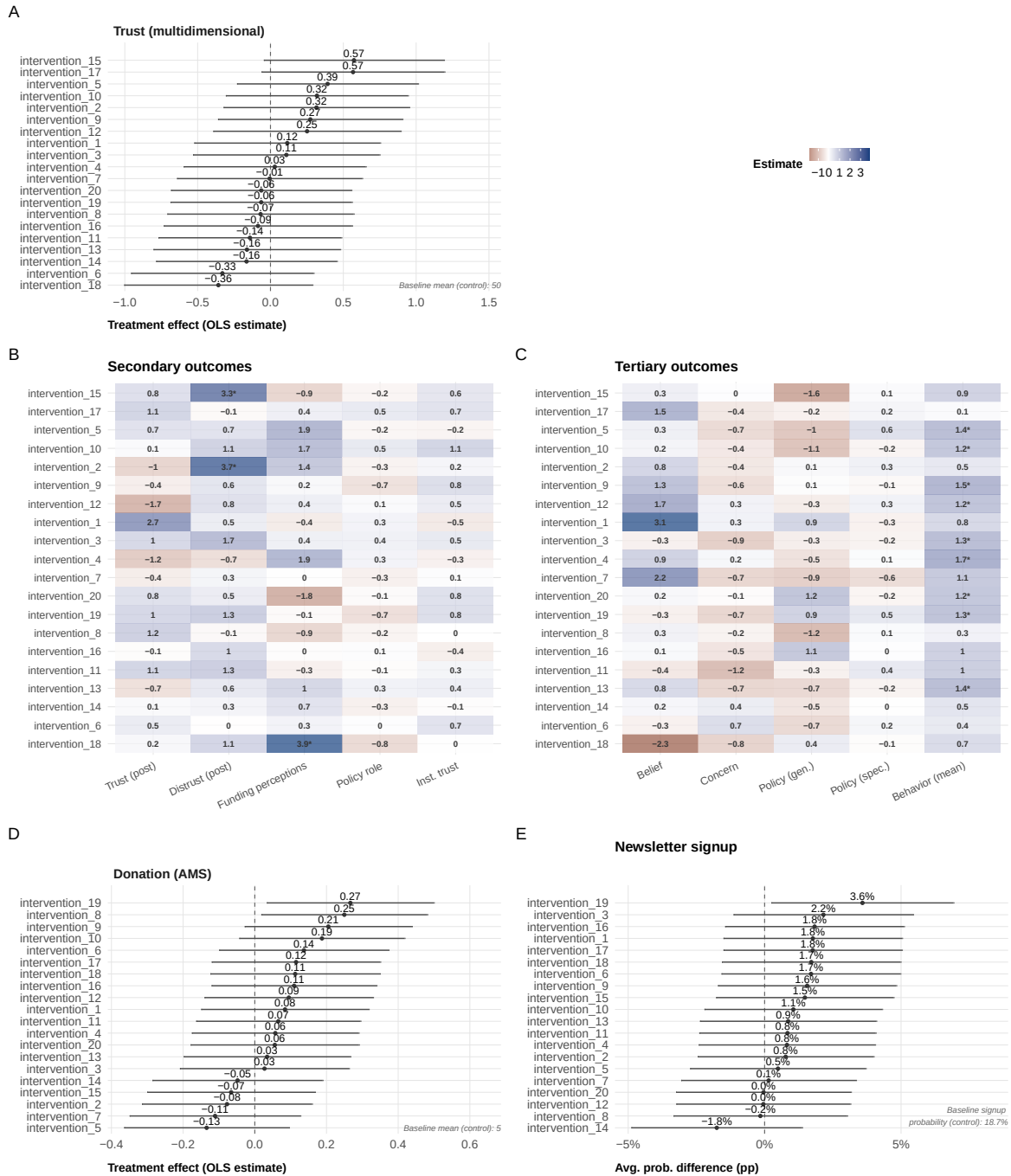


Figure 3: Overview of treatment effects across all outcomes. **A** Estimated treatment effects on the primary outcome (trust in climate scientists, multidimensional scale), ordered from largest to smallest effect. **B–C** Treatment effects as heatmaps for secondary and tertiary outcomes, with interventions in the same order as panel A; cell labels show the estimate and BH-adjusted significance stars. **D** Treatment effects on donation to the American Meteorological Society (continuous, 0–10 scale). **E** Treatment effects on newsletter signup (binary outcome), reported as predicted probabilities from logistic regression. Whiskers in panels A, D, and E depict 95% CIs without correction for multiple comparisons. In all panels, asterisks indicate BH-adjusted significance: * $p_{adj} < .05$; ** $p_{adj} < .01$; *** $p_{adj} < .001$. Data are simulated at random, thus we should not expect to see any significant effects.

Attrition and missing values

We define attrition as a case where a participant does not respond to an outcome measure. There are two cases of attrition: First, a participant drops out of the survey, i.e. does not finish it. We allow for that at any time of the survey. Second, a participant completes the survey, but does not answer all questions. This is possible, as we do not force responses, with the exception of quota relevant variables.

The above definition of attrition is outcome based: A participant who has missing values for one or multiple outcome measures will still be included in the analyses on all outcome measures for which they provided data. For example, a participant might have answered the main multi-dimensional trust measure and will be considered a complete case for all analyses regarding this variable. But the same participant might not have answered the donation outcome question, and will be treated as a missing value for all analyses regarding that variable. We will report missing values for all key variables, along with other descriptive statistics (see Table 4 for an example).

Running a study on a large sample with many experimental conditions, it is likely that we will face the issue of differential attrition—when, after treatment assignment, the attrition rate differs systematically between experimental conditions. Differential attrition can bias estimates of treatment effects. To illustrate, consider the following scenario: Some interventions might require more effort from participants than others (e.g., interacting with a chatbot vs. reading a short text). Participants who are generally not willing to make much of an effort might drop out of high-effort treatment conditions, but not the low-effort conditions. Suppose that, in general, those participants who are not willing to make an effort also tend to trust climate scientists less. Now, these participants would drop out in the high-effort conditions, but not in the low-effort ones. As a consequence, a naive estimate of the treatment effect for the high-effort conditions will be overestimated—all the low-trust participants who were not willing to make an effort dropped out and do not count into the high-effort conditions average, while they do count into the average of low-effort conditions.

Tests for differential attrition

To test for differential attrition, we follow procedures established in prior megastudies (Voelkel et al. 2024, 2026). We implement two complementary tests.

First, we estimate whether the number of missing responses differs between conditions. We run a linear probability model in which a binary indicator for study completion is regressed on experimental condition. We then conduct a heteroskedasticity-robust F-test of the joint hypothesis that attrition rates in all treatment conditions equal the attrition rate in the control condition.


```

run_attrition_f_test <- function(data,
                                outcome,
                                condition_var = "condition") {

  # Completion indicator for the specific outcome
  model_data <- data %>%
    mutate(
      completed = if_else(
        is.na(.data[[outcome]]),
        FALSE,
        TRUE,
      ),
      completed_numeric = as.numeric(completed)
    )

  # skip test if no variation (i.e. if everyone completed/attrited)
  if(length(unique(model_data$completed)) < 2){
    return(tibble(outcome = outcome, Chi2 = NA_real_, p_value = NA_real_))
  }

  formula <- as.formula(paste("completed ~", condition_var))

  model <- lm(formula, data = model_data)

  # Only the coefficients for the condition variable (not the intercept)
  test_terms <- grep(condition_var, names(model$coefficients), value = TRUE)

  f_test <- car::linearHypothesis(model, test_terms, white.adjust = "hc2")

  tibble(
    outcome = outcome,
    F_statistic = f_test$F[2],
    p_value = f_test$`Pr(>F)`[2]
  )
}

# Run attrition test 1: Condition only
attrition_f_results <- map_df(
  outcomes,
  ~ run_attrition_f_test(
    data = data,
    outcome = .x
  )
)

```

```

    )
  )

# check
# attrition_f_results

```

Second, we test whether characteristics of participants with missing values differ between conditions (heterogenous attrition). This second test is important, because even if overall attrition rates are similar, the composition of who drops out could be affected by treatment assignment. For this test, we add to the linear probability model from the first test a set of covariates and their interactions with experimental condition. These covariates will be the same we will later use to account for differential attrition (if necessary). Their selection process is described in the next section. We again conduct a heteroskedasticity-robust F-test, this time testing whether all condition-by-covariate interaction terms are jointly equal to zero.

```

run_attrition_interactions <- function(data,
                                       outcome,
                                       condition_var = "condition",
                                       covariates) {

  # Completion indicator
  model_data <- data %>%
    mutate(
      completed = if_else(
        is.na(.data[[outcome]]),
        FALSE,
        TRUE,
      ),
      completed_numeric = as.numeric(completed)
    )

  # Skip if no variation
  if(length(unique(model_data$completed)) < 2){
    return(tibble(outcome = outcome,
                  covariate = covariates,
                  F_statistic = NA_real_,
                  p_value = NA_real_))
  }

  # Loop over covariates
  interaction_tests <- covariates %>%
    map_df(function(cov) {

```

```

# Build formula for condition * covariate
formula <- as.formula(paste0("completed ~ ", condition_var, " * ", cov))
model <- lm(formula, data = model_data)

# Identify interaction terms (condition:covariate)
interaction_terms <- grep(":", names(coef(model)), value = TRUE)

# Skip if no interaction terms
if(length(interaction_terms) == 0){
  return(tibble(outcome = outcome,
                 covariate = cov,
                 F_statistic = NA_real_,
                 p_value = NA_real_))
}

# Joint F-test with robust SE
f_test <- car::linearHypothesis(model,
                                interaction_terms,
                                white.adjust = "hc1")

tibble(
  outcome = outcome,
  covariate = cov,
  F_statistic = f_test$F[2],
  p_value = f_test$`Pr(>F)`[2]
)

}) |>
# adjust for multiple comparison
mutate(adjusted_p.value = p.adjust(p_value, method = "BH"))

return(interaction_tests)
}

# Run attrition test 2: Condition × Covariates
attrition_interaction_results <- map_df(
  outcomes_illustrative,
  ~ run_attrition_interactions(
    data = data,
    outcome = .x,
    covariates = covariates
  )
)

```

```
)  
  
# check  
# attrition_interaction_results
```

Account for differential attrition

In line with other megastudies (Voelkel et al. 2024, 2026), if we find evidence of heterogeneous differential attrition, we will use inverse-probability weighting (IPW) for all our analyses.

IPW adjusts the analysis by upweighting participants who completed the study but resemble those who dropped out (or, more generally, have missing values for a particular outcome), based on their observed characteristics. Specifically, we model the probability of completing the study as a function of a set of pre-treatment covariates using a random forest classifier. We use a random forest because it flexibly captures nonlinear relationships and interactions between predictors without requiring model specification decisions. The predicted completion probability for each participant is then used to compute inverse probability weights, which are passed to the weighted regression models.

IPW relies on one key assumption: conditional on the observed covariates included in the weighting model, attrition is independent of participants' potential outcomes. In other words, after accounting for measured pre-treatment characteristics, whether a participant drops out is unrelated to what their outcome would have been. This assumption implies that all systematic predictors of attrition that are also related to the outcome must be observed and included in the weighting model. If attrition depends on unmeasured factors that also affect the outcome, IPW cannot fully eliminate bias. While this assumption cannot be tested directly and may not hold perfectly in practice, including a broad set of pre-treatment covariates in the weighting model reduces the risk of residual confounding. We therefore interpret IPW-adjusted estimates as reducing—but not necessarily eliminating—concerns about bias due to differential attrition.

However, there is a trade-off in how many covariates to include in the IPW weighting model. On the one hand, IPW is less biased when based on more predictor variables. On the other hand, covariates with missing values pose a practical challenge: a weighting model that relies on complete cases only assigns weights to the subset of participants with valid responses on all covariates. Participants without weights are excluded from IPW-weighted analyses—though they remain included in the unweighted analyses. The tradeoff is thus between more accurate weights based on a larger covariate set but estimated in a reduced sample, versus less accurate weights based only on fully observed variables (i.e., the quota-relevant variables gender, age, and race) but estimated in the full sample.

We resolve this trade-off as follows. As a baseline, we always include condition, gender, age, and race, as these are mandatory and have no missing values. Beyond these, we include up to three additional pre-treatment variables—single-item trust in climate scientists, partisan

identity, and education level—provided their individual missingness rate does not exceed 5% in the final sample⁶. We expect these three additional variables to be related both to attrition and our outcome variables. We cap the number of additional variables at three to limit maximum sample loss from listwise deletion to approximately 15%. In practice, sample loss is likely lower than this upper bound, as missingness across optional questions tends to be correlated—participants who skip one question tend to skip others, too. Should any of the additional variables exceed the 5% threshold, we will exclude them from the weighting model.

```
# baseline predictors (mandatory, always included)
baseline_predictors <- c("gender", "age", "race")

# candidate additional predictors
candidate_predictors <- c("trust_pre", "party", "education")

# check missingness rates
missingness <- data |>
  summarise(across(all_of(candidate_predictors),
    ~ mean(is.na(.x)))) |>
  pivot_longer(everything(),
    names_to = "variable",
    values_to = "missingness_rate")

# build final weight predictor list
additional_predictors <- missingness |>
  filter(missingness_rate < 0.05) |>
  pull(variable)

weight_predictors <- c(baseline_predictors, additional_predictors)
```

Table 3: Overview of missingness rates for IPW predictor candidate variables.

variable	missing- ness_pct	in- cluded_in_ipw
trust_pre	0.0%	Yes
party	0.0%	Yes
education	0.0%	Yes

⁶Note that in the simulated data used in this pre-registration, no pre-treatment variables have missing values, see Table 3

```

get_ipw_weights_rf <- function(data,
                               outcome,
                               condition_var = "condition",
                               weight_predictors,
                               ntree = 200) {

  # Completion indicator
  dat <- data |>
    mutate(
      completed = factor(
        !is.na(.data[[outcome]]),
        levels = c(FALSE, TRUE),
        labels = c("no", "yes")
      )
    )

  # Build formula explicitly
  predictors <- c(condition_var, weight_predictors)
  rf_formula <- as.formula(
    paste("completed ~", paste(predictors, collapse = " + "))
  )

  # Fit random forest
  rf_model <- randomForest::randomForest(
    formula = rf_formula,
    data = dat,
    importance = TRUE,
    ntree = ntree,
    na.action = na.exclude # safety net for any remaining NAs
  )

  # Predicted probability of completion
  p_complete <- predict(rf_model, newdata = dat, type = "prob")[, "yes"]

  # Inverse probability weights + trimming at 99th percentile
  dat <- dat |>
    mutate(
      p_complete = p_complete,
      ipw = 1 / p_complete,
      ipw_trimmed = pmin(ipw, quantile(ipw, 0.99))
    )
}

```

```

    return(dat)
  }

# set a seed to make random forest procedure reproducible
set.seed(28367)

run_all_outcomes_ipw <- function(data,
                                outcomes,
                                condition_var = "condition",
                                weight_predictors,
                                covariates = NULL) {

  purrr::map_df(outcomes, function(outcome) {

    # --- Unweighted main model
    main_unweighted <- run_main_treatment_model(
      data = data,
      outcome = outcome,
      condition_var = condition_var,
      covariates = covariates
    ) %>%
      mutate(model = "Unweighted")

    # --- IPW weights via random forest
    dat_ipw <- get_ipw_weights_rf(
      data = data,
      outcome = outcome,
      condition_var = condition_var,
      weight_predictors = weight_predictors
    )

    # --- Weighted robustness model
    main_weighted <- run_main_treatment_model(
      data = dat_ipw,
      outcome = outcome,
      condition_var = condition_var,
      covariates = covariates,
      weights = "ipw"
    ) %>%
      mutate(model = "IPW")

    bind_rows(main_unweighted, main_weighted)
  })
}

```

```

  })
}

# run robustness analysis that compares ipw and unweighted
results_ipw <- run_all_outcomes_ipw(
  data = data,
  outcomes = outcomes_illustrative,
  weight_predictors = weight_predictors,
  covariates = covariates
)

```

If we use IPW due to differential attrition, we will also report how it compares to results without using IPW in the supplemental materials (see [Figure 9](#) for a possible illustration).

Moderators

We will examine whether the effects of the interventions vary as a function of a set of moderator variables assessed prior to treatment, including demographic variables, political identity, religion, and belief in climate change. Specifically, our moderator variables are:

- age
- gender
- race
- education
- income
- social_class
- urban_rural
- party
- religion
- born_again
- religiosity
- belief_pre
- trust_pre

Moderator analyses will be conducted separately for each moderator and each outcome. To estimate moderator effects, we will add the moderator variable as an interaction term to the OLS regression used to assess the main treatment effects. We will not add any covariates. As for the main treatment effect, we use heteroskedasticity-robust standard errors. We account for multiple comparisons using the Benjamini–Hochberg procedure, applied separately within each combination of moderator and outcome. For continuous moderators, p-values are adjusted

across the 20 intervention-specific slopes; for categorical moderators, p-values are adjusted across all interaction terms (20 interventions $\times$ number of moderator levels minus one).

Formally, for a given outcome Y_i and moderator M_i , we estimate the following model:

$$Y_i = \beta_0 + \sum_{k=1}^K \beta_k D_{ik} + \delta M_i + \sum_{k=1}^K \theta_k (D_{ik} \times M_i) + \varepsilon_i,$$

where:

- Y_i is the outcome variable for participant i (e.g., trust in climate scientists).
- β_0 is the intercept, representing the expected outcome in the main control condition when all covariates and the moderator equal zero.
- D_{ik} is a dummy variable equal to 1 if participant i was assigned to intervention k , and 0 otherwise. The main control condition serves as the omitted reference category, and the interactive control condition is excluded from the estimation sample.
- β_k captures the average effect of intervention k relative to the control condition when the moderator equals zero.
- M_i is the moderator variable of interest.
- δ captures the association between the moderator and the outcome in the control condition.
- $D_{ik} \times M_i$ denotes the interaction between intervention k and the moderator.
- θ_k captures how the effect of intervention k changes as a function of the moderator.
- γ is the corresponding vector of coefficients for the covariates.
- ε_i is an error term capturing unexplained variation in the outcome.

```
run_moderator_model <- function(data,
                                outcome,
                                moderator,
                                condition_var = "condition",
                                covariates = NULL,
                                weights = NULL,
                                adjust_method = "BH") {

  rhs <- paste(c(paste0(condition_var, " * ", moderator), covariates),
              collapse = " + ")
  model_formula <- as.formula(paste(outcome, "~", rhs))
  baseline <- levels(data[[condition_var]])[1]

  fit <- lm(
    model_formula,
    data = data,
    weights = if (!is.null(weights)) data[[weights]] else NULL
```

```

)

vcov_robust <- sandwich::vcovHC(fit, type = "HC2")

interaction_effects <- lmtest::coeftest(fit, vcov = vcov_robust) |>
  broom::tidy(conf.int = TRUE) |>
  filter(str_detect(term, ":")) |>
  mutate(
    baseline          = baseline,
    condition          = str_extract(term, paste0("(?<=", condition_var, ")")["^:]+")),
    moderator_level    = str_remove(str_extract(term, "(?<=:).+"), moderator),
    p.value_adjusted    = p.adjust(p.value, method = adjust_method),
    significant_adjusted = case_when(
      p.value_adjusted < .001 ~ "****",
      p.value_adjusted < .01  ~ "***",
      p.value_adjusted < .05  ~ "**",
      TRUE                   ~ NA_character_
    )
  )

is_numeric_mod <- is.numeric(data[[moderator]])

if (!is_numeric_mod) {
  predicted_effects <- margaleffects::avg_comparisons(
    fit,
    variables = condition_var,
    by        = moderator,
    vcov      = vcov_robust,
    newdata   = "mean"
  ) |>
  as_tibble() |>
  mutate(
    condition          = str_remove(contrast, " - .+$"),
    moderator_level    = .data[[moderator]],
    baseline          = baseline,
    p.value_adjusted    = p.adjust(p.value, method = adjust_method),
    significant_adjusted = case_when(
      p.value_adjusted < .001 ~ "****",
      p.value_adjusted < .01  ~ "***",
      p.value_adjusted < .05  ~ "**",
      TRUE                   ~ NA_character_
    )
  )
}

```

```

    ) |>
    filter(!is.na(condition)) |>
    select(condition, moderator_level, estimate, conf.low, conf.high,
           p.value, p.value_adjusted, significant_adjusted, baseline)
  }

  if (is_numeric_mod) {
    predicted_effects <- marginaleffects::comparisons(
      fit,
      variables = condition_var,
      vcov       = vcov_robust,
      newdata    = do.call(
        marginaleffects::datagrid,
        c(list(model = fit),
          setNames(list(fivenum(data[[moderator]])), moderator))
      )
    ) |>
    as_tibble() |>
    mutate(
      condition          = str_remove(contrast, " - .+$"),
      moderator_value    = .data[[moderator]],
      baseline           = baseline,
      p.value_adjusted   = p.adjust(p.value, method = adjust_method),
      significant_adjusted = case_when(
        p.value_adjusted < .001 ~ "****",
        p.value_adjusted < .01  ~ "***",
        p.value_adjusted < .05  ~ "**",
        TRUE                  ~ NA_character_
      )
    ) |>
    filter(!is.na(condition)) |>
    select(condition, moderator_value, estimate, conf.low, conf.high,
           p.value, p.value_adjusted, significant_adjusted, baseline)
  }

  list(
    interaction_effects = interaction_effects,
    predicted_effects   = predicted_effects
  )
}

```

```

# run moderator models for all outcomes × moderators
moderator_results_list <- expand_grid(
  outcome = outcomes_illustrative,
  moderator = moderators_illustrative
) |>
mutate(
  results = map2(
    outcome, moderator,
    ~ run_moderator_model(
      data = data,
      outcome = .x,
      moderator = .y,
      covariates = NULL
    )
  )
)

# extract interaction effects
moderator_results <- moderator_results_list |>
mutate(results = map(results, "interaction_effects")) |>
unnest(results)

# extract predicted effects
moderator_results_predicted <- moderator_results_list |>
mutate(results = map(results, "predicted_effects")) |>
unnest(results)

```

In the manuscript, we will focus on moderator effects regarding our main outcome, trust in climate scientists. In the supplemental material, we will report moderator effects on all secondary and tertiary outcomes. For categorical moderators, we will visualize the predicted treatment effects per category in the manuscript (Figure 4). For example, for gender, we will visualize the predicted treatment effect for men, women, and other. We will report whether the differences between these categories—the moderator effect, or interaction term from the model—are significant. For continuous moderators, we will visualize the interaction term, i.e. the estimated changes in the treatment effects per unit increase in the moderator (Figure 5). We will provide detailed tables of the interaction terms for all moderators on all outcomes in the supplemental materials (see, e.g., Table 6).

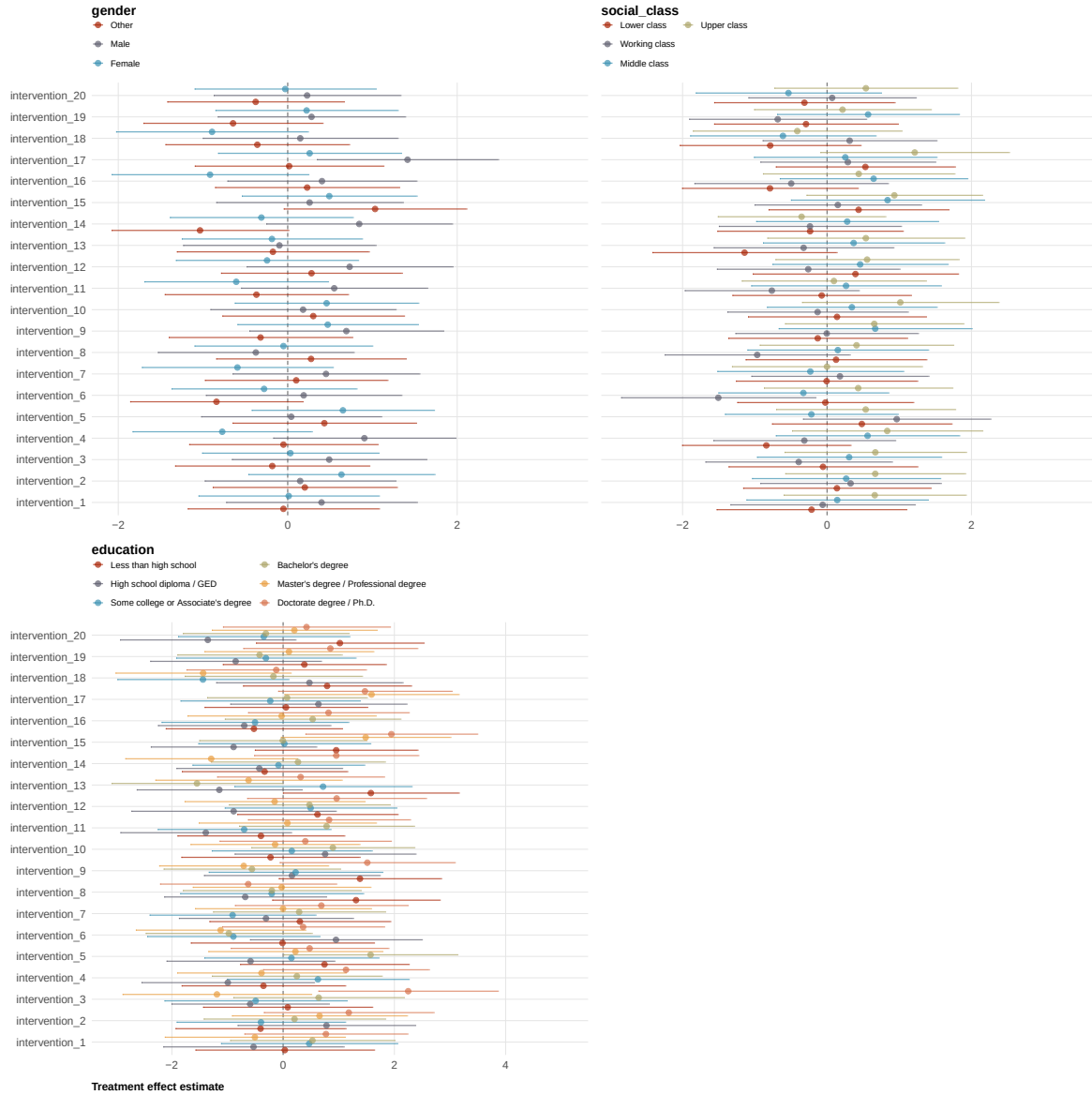


Figure 4: Predicted treatment effects by categorical moderators. Each panel shows the estimated treatment effect within each moderator category, on the outcome scale. Dots represent predicted effects, whiskers depict 95% CIs without correction for multiple comparisons. Significance stars indicate whether the treatment effect in a given category differs significantly from zero, based on BH-adjusted p-values across 20 interventions $\times$ n_levels comparisons within each outcome. Data are simulated at random, thus we should not expect to see any significant effects.

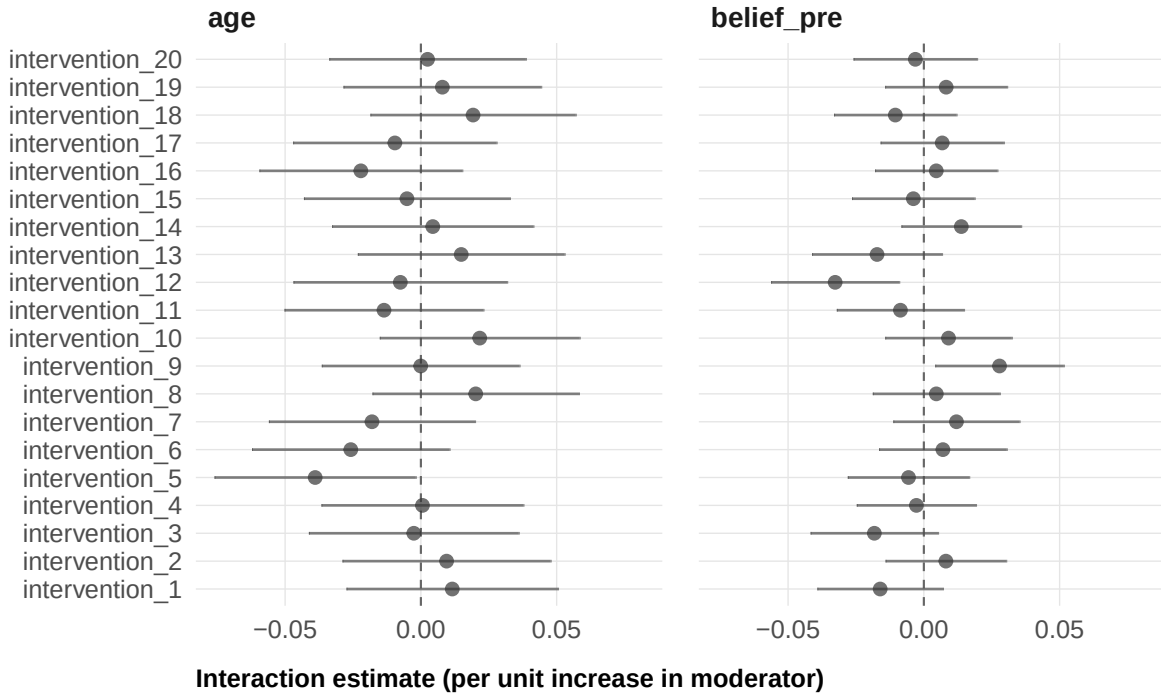


Figure 5: Moderation of treatment effects by continuous moderators. Dots represent estimated changes in the treatment effects per unit increase in the moderator, whiskers depict 95% CIs without correction for multiple comparison. A positive estimate indicates the intervention is more effective for participants with higher values of the moderator. Asterisks indicate significance after BH adjustment: * $p_{adj} < .05$; ** $p_{adj} < .01$; *** $p_{adj} < .001$. Data are simulated at random, thus we should not expect to see any significant effects.

Partisan identity and importance

Given the well-documented partisan gap in trust in climate scientists and attitudes toward science more broadly, we treat partisan identity as a separate case in our moderator analyses. We conduct two related analyses. First, we examine whether treatment effects differ across party lines (in the same way, as for other categorical moderator variables). Second, we examine the role of *partisan importance*—how important being a supporter of a party is to participants. We expect this variable to operate differently across parties: among Republicans, stronger partisan identity likely reinforces skepticism toward scientific institutions, potentially dampening intervention effects; among Democrats, the same identification may reinforce receptivity to pro-science messaging. We therefore estimate the partisan importance model separately for Republicans and Democrats, allowing us to test whether the strength of partisan cue-taking within each party moderates intervention effectiveness.

For the partisan importance analysis, although models are estimated on separate subsamples, we treat both as part of a single conceptual family of tests. BH adjustment is therefore applied jointly across both subsamples — that is, across 40 comparisons (20 interventions $\times$ 2 parties) per outcome — resulting in more conservative p-values than within-party adjustment alone would produce.

```
# partisan importance moderator - run separately for Republicans and Democrats

# run models for both parties
moderator_results_partisan_importance <- expand_grid(
  outcome = outcomes_illustrative,
  party    = c("Republican", "Democrat")
) |>
mutate(
  results = map2(
    outcome, party,
    ~ run_moderator_model(
      data      = data |> filter(party == .y),
      outcome   = .x,
      moderator = "party_importance",
      covariates = covariates
    )$interaction_effects
  )
) |>
unnest(results) |>
# override p.value_adjusted: BH across both parties jointly
# family = 40 comparisons (20 interventions  $\times$  2 parties) per outcome
group_by(outcome) |>
mutate(
  p.value_adjusted = p.adjust(p.value, method = "BH"),
  significant_adjusted = case_when(
    p.value_adjusted < .001 ~ "***",
    p.value_adjusted < .01  ~ "**",
    p.value_adjusted < .05  ~ "*",
    TRUE                    ~ NA_character_
  )
) |>
ungroup()
```

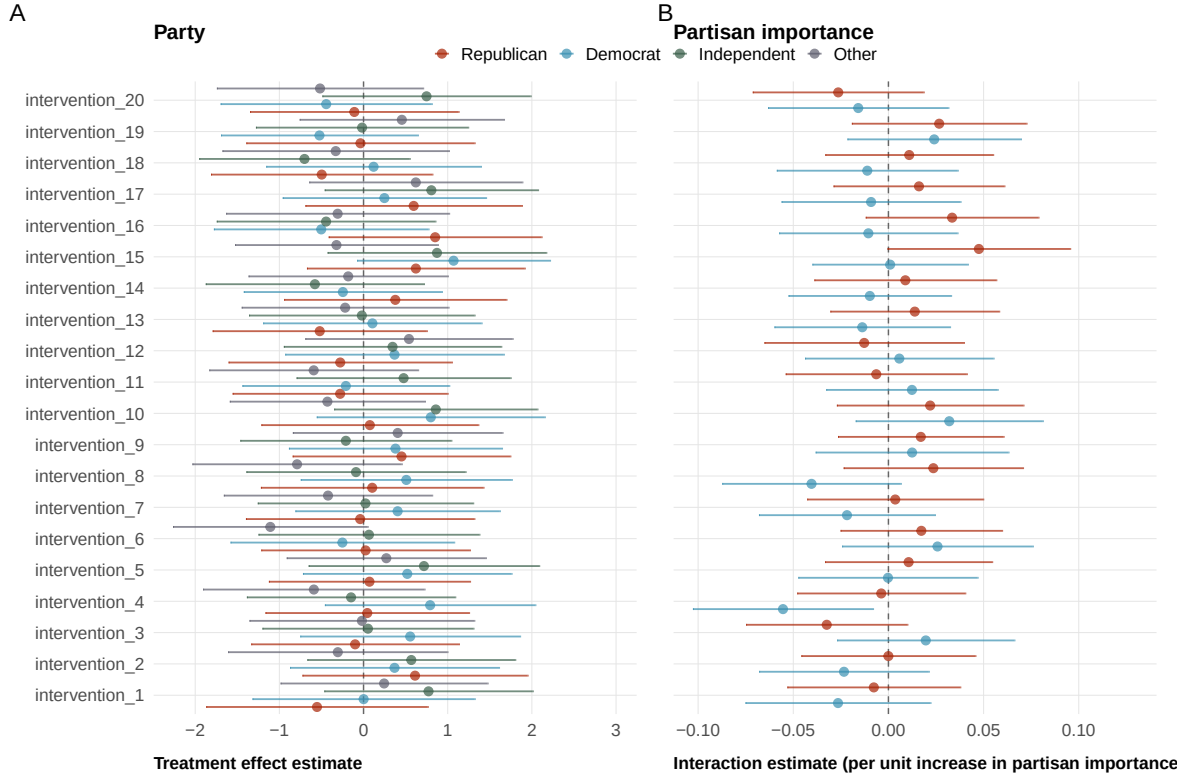


Figure 6: Moderation of treatment effects by party identification and partisan importance. **A** Predicted treatment effects within each partisan identity category. Significance stars indicate whether the treatment effect differs significantly from zero, based on BH-adjusted p-values across 60 comparisons (20 interventions $\times$ 3 party categories) within each outcome. **B** Interaction estimates for partisan importance (0–100 slider) estimated separately for Republicans and Democrats, showing how the treatment effect changes per unit increase in partisan importance within each party. Significance stars based on BH-adjusted p-values across 40 comparisons (20 interventions $\times$ 2 parties) within each outcome. Whiskers depict 95% CIs without correction for multiple comparisons. Data are simulated at random, thus we should not expect to see any significant effects.

Binary outcome: newsletter signup

Moderator analyses for newsletter signup follow the same structure as for the continuous outcomes described above, with treatment $\times$ moderator interaction terms added to the baseline logistic regression described in the treatment effect section. We report interaction effects as odds ratios, together with the other moderator effects in the Appendix (see, e.g., Table 9).

Odds ratios above 1 indicate that the intervention is more effective for that group relative to the reference category. For better interpretation, we will also report predicted treatment effects on the probability scale. BH adjustment follows the same logic as for continuous outcomes: applied within each moderator $\times$ outcome combination across all 20 intervention-vs-control comparisons for interaction terms, and across 20 interventions $\times$ n_levels comparisons for predicted effects.

```
run_moderator_model_binary <- function(data,
                                     outcome,
                                     moderator,
                                     condition_var = "condition",
                                     covariates     = NULL,
                                     weights         = NULL,
                                     adjust_method  = "BH") {

  rhs      <- paste(c(paste0(condition_var, " * ", moderator), covariates),
                    collapse = " + ")
  model_formula <- as.formula(paste(outcome, "~", rhs))
  baseline     <- levels(data[[condition_var]])[1]

  fit <- glm(
    model_formula,
    data      = data,
    family    = binomial(link = "logit"),
    weights   = if (!is.null(weights)) data[[weights]] else NULL
  )

  vcov_robust <- sandwich::vcovHC(fit, type = "HC2")
  is_numeric_mod <- is.numeric(data[[moderator]])

  interaction_effects <- lmtest::coeftest(fit, vcov = vcov_robust) |>
    broom::tidy(conf.int = TRUE) |>
    filter(str_detect(term, ":")) |>
    mutate(
      baseline           = baseline,
      condition          = str_extract(term, paste0("(?<=", condition_var, ")") [^:]+)),
      moderator_level    = str_remove(str_extract(term, "(?<=:).+"), moderator),
      p.value_adjusted    = p.adjust(p.value, method = adjust_method),
      significant_adjusted = case_when(
        p.value_adjusted < .001 ~ "****",
        p.value_adjusted < .01  ~ "***",
        p.value_adjusted < .05  ~ "**",
        TRUE                  ~ NA_character_
      )
    )
}
```

```

    ),
    odds_ratio = exp(estimate),
    or_conf.low = exp(conf.low),
    or_conf.high = exp(conf.high)
  ) |>
  select(-term)

if (!is_numeric_mod) {
  predicted_effects <- margaleffects::avg_comparisons(
    fit,
    variables = condition_var,
    by = moderator,
    vcov = vcov_robust,
    newdata = "mean"
  ) |>
  as_tibble() |>
  mutate(
    condition = str_remove(contrast, " - .+$"),
    moderator_level = .data[[moderator]],
    baseline = baseline
  ) |>
  filter(!is.na(condition)) |>
  select(condition, moderator_level, estimate, conf.low, conf.high,
    p.value, baseline)
}

if (is_numeric_mod) {
  predicted_effects <- margaleffects::comparisons(
    fit,
    variables = condition_var,
    vcov = vcov_robust,
    newdata = do.call(
      margaleffects::datagrid,
      c(list(model = fit),
        setNames(list(fivenum(data[[moderator]])), moderator))
    )
  ) |>
  as_tibble() |>
  mutate(
    condition = str_remove(contrast, " - .+$"),
    moderator_value = .data[[moderator]],
    baseline = baseline
  )
}

```

```

    ) |>
    filter(!is.na(condition)) |>
    select(condition, moderator_value, estimate, conf.low, conf.high,
           p.value, baseline)
  }

  list(
    interaction_effects = interaction_effects,
    predicted_effects   = predicted_effects
  )
}

# run moderator models for the binary outcome × moderators
moderator_results_binary_list <- expand_grid(
  outcome = "newsletter_signup",
  moderator = moderators_illustrative
) |>
mutate(
  results = map2(
    outcome, moderator,
    ~ run_moderator_model_binary(
      data      = data,
      outcome   = .x,
      moderator = .y,
      covariates = NULL
    )
  )
)

moderator_results_binary <- moderator_results_binary_list |>
mutate(results = map(results, "interaction_effects")) |>
unnest(results)

moderator_results_binary_predicted <- moderator_results_binary_list |>
mutate(results = map(results, "predicted_effects")) |>
unnest(results)

```

Persistence

We will test whether treatment effects observed in the experiment persist with a follow-up survey, fielded one week after the end of data collection of the experiment. We will run two

tests of persistence: First, we test whether treatment effects remain present in the follow-up survey. This corresponds to running the same model as for the main treatment effects in the experiment, but on the data from the follow-up survey. Second, we test whether the follow-up effects are statistically different from the main survey effects.

```
# combine surveys

data_followup <- data_followup |>
  mutate(time = "follow_up") |>
  left_join(data |> select(id, condition), by = "id")

followup_conditions <- unique(data_followup$condition)

data_reduced <- data |>
  filter(condition %in% followup_conditions) |>
  droplevels() |>
  mutate(time = "experiment")

data_followup <- data_reduced |>
  select(id,
         condition,
         all_of(demographics),
         all_of(covariates),
         all_of(moderators),
         ) |>
  left_join(
    data_followup |> select(-condition), # drop condition from data_followup
    by = "id"
  )

# merge data sets (reduced conditions)
merged_data <- data_reduced |>
  bind_rows(data_followup) |>
  mutate(
    time = relevel(factor(time), ref = "experiment")
  )
```

We can run both tests from the a single interaction model, combining the data from the experiment and the follow-up survey. We will stack the main and follow-up survey data into a long-format panel dataset and estimate linear regression models including a treatment $\times$ wave interaction term. Standard errors will be clustered at the participant level to account for repeated observations across waves. We will use the same covariates (age, race, gender) and the same adjustment for multiple comparison as for the main treatment effect model.

Formally, our model for testing persistence is:

$$Y_{it} = \beta_0 + \sum_{k=1}^K \beta_k D_{ik} + \beta_T Time_t + \sum_{k=1}^K \beta_{kT} (D_{ik} \times Time_t) + \gamma' \mathbf{X}_i + \varepsilon_{it}$$

where

- Y_{it} is the outcome for participant i at time t (experiment = 0, follow-up = 1).
- D_{ik} are dummies for intervention k , with the main control condition as reference.
- $Time_t$ indicates experiment (0) vs. follow-up (1).
- $\mathbf{X}_i$ = vector of pre-treatment covariates (gender, age, race), with corresponding coefficient vector γ .
- β_k = treatment effect in the experiment sample, for the reduced sample of participants who also completed the follow-up.
- β_{kT} = change in effect; statistical test of persistence.
- $\beta_k + \beta_{kT}$ = treatment effect in the follow-up sample.
- ε_{it} = error term, with standard errors clustered at the participant level to account for repeated observations across waves.

As in the experiment, we will test for attrition, and for differential attrition, in the follow-up survey. Note that we will not run persistence analyses for newsletter signup, as this variable is not included in the follow-up survey.

```
# Check baseline condition
# levels(data_followup$condition)

# Run attrition test 1: Condition only
attrition_f_results <- map_df(
  outcomes_illustrative,
  ~ run_attrition_f_test(
    data = data_followup,
    outcome = .x
  )
)

# check
# attrition_f_results
```

```
# Run attrition test 2: Condition × Covariates
attrition_interaction_results <- map_df(
  outcomes_illustrative,
  ~ run_attrition_interactions(
```

```

    data = data_followup,
    outcome = .x,
    covariates = covariates
  )
)

# check
# attrition_interaction_results

```

Should we find evidence of differential attrition, we will address it using inverse probability of retention weights (IPW), estimated separately for the follow-up survey. We will use the same random forest approach, with the same weight predictors, as described above for the experiment. The data from the follow-up survey will contain all participants from the experiment, with NA on all follow-up outcomes if they did not respond. Therefore, our completion estimate not only captures who completed the follow-up survey, but also who took to the follow-up survey at all. We estimate IPW separately for experiment and follow-up because attrition patterns may differ between two, and IPW is intended to address these different patterns. As a result, a participant who completed both the experiment and the follow-up survey may receive different weights in the two data sets.

```

run_persistence_model <- function(data,
                                  outcome,
                                  condition_var = "condition",
                                  covariates = NULL,
                                  weights = NULL,
                                  id_var = "id",
                                  time_var = "time",
                                  adjust_method = "BH") {

  # Formula: condition × time interaction
  rhs <- paste(c(paste0(condition_var, " * ", time_var),
                 covariates),
              collapse = " + ")
  model_formula <- as.formula(paste(outcome, "~", rhs))

  # Baseline (control) level
  baseline <- levels(data[[condition_var]])[1]

  # Fit
  fit <- lm(
    model_formula,
    data = data,

```

```

  weights = if (!is.null(weights)) data[[weights]] else NULL
)

# Cluster-robust VCOV at participant level
vcov_clustered <- sandwich::vcovCL(fit,
                                   cluster = as.formula(paste0("~", id_var)
                                   )
)

# Interaction terms (condition × time)
interaction_effects <- lmtest::coeftest(fit, vcov = vcov_clustered) |>
  broom::tidy(conf.int = TRUE) |>
  filter(str_detect(term, ":")) |>
  mutate(
    baseline          = baseline,
    outcome            = outcome,
    condition          = str_extract(term, paste0("(?<=", condition_var, ")[^:]+")),
    p.value_adjusted   = p.adjust(p.value, method = adjust_method),
    significant_adjusted = case_when(
      p.value_adjusted < .001 ~ "***",
      p.value_adjusted < .01  ~ "**",
      p.value_adjusted < .05  ~ "*",
      TRUE                   ~ NA_character_
    )
  )

# Predicted effects within each wave
predicted_effects <- margaleffects::avg_comparisons(
  fit,
  variables = condition_var,
  by        = time_var,
  vcov      = vcov_clustered,
  newdata   = "mean"
) |>
  as_tibble() |>
  mutate(
    condition          = str_remove(contrast, " - .+$"),
    baseline           = baseline,
    outcome             = outcome,
    p.value_adjusted    = p.adjust(p.value, method = adjust_method),
    significant_adjusted = case_when(
      p.value_adjusted < .001 ~ "***",

```

```

      p.value_adjusted < .01 ~ "**",
      p.value_adjusted < .05 ~ "*",
      TRUE ~ NA_character_
    )
  ) |>
  filter(!is.na(condition)) |>
  select(condition, !!time_var := .data[[time_var]],
         estimate, conf.low, conf.high,
         p.value, p.value_adjusted, significant_adjusted,
         baseline, outcome)

  list(
    interaction_effects = interaction_effects,
    predicted_effects   = predicted_effects
  )
}

```

```

# run persistence model on all continuous outcomes

followup_results_list <- map(
  outcomes_illustrative,
  ~ run_persistence_model(
    data      = merged_data,
    outcome   = .x,
    covariates = covariates
  )
)

followup_results_predicted <- map_df(followup_results_list,
                                     "predicted_effects") |>
  filter(condition %in% followup_conditions)

followup_results_interaction <- map_df(followup_results_list,
                                       "interaction_effects") |>
  filter(condition %in% followup_conditions)

```

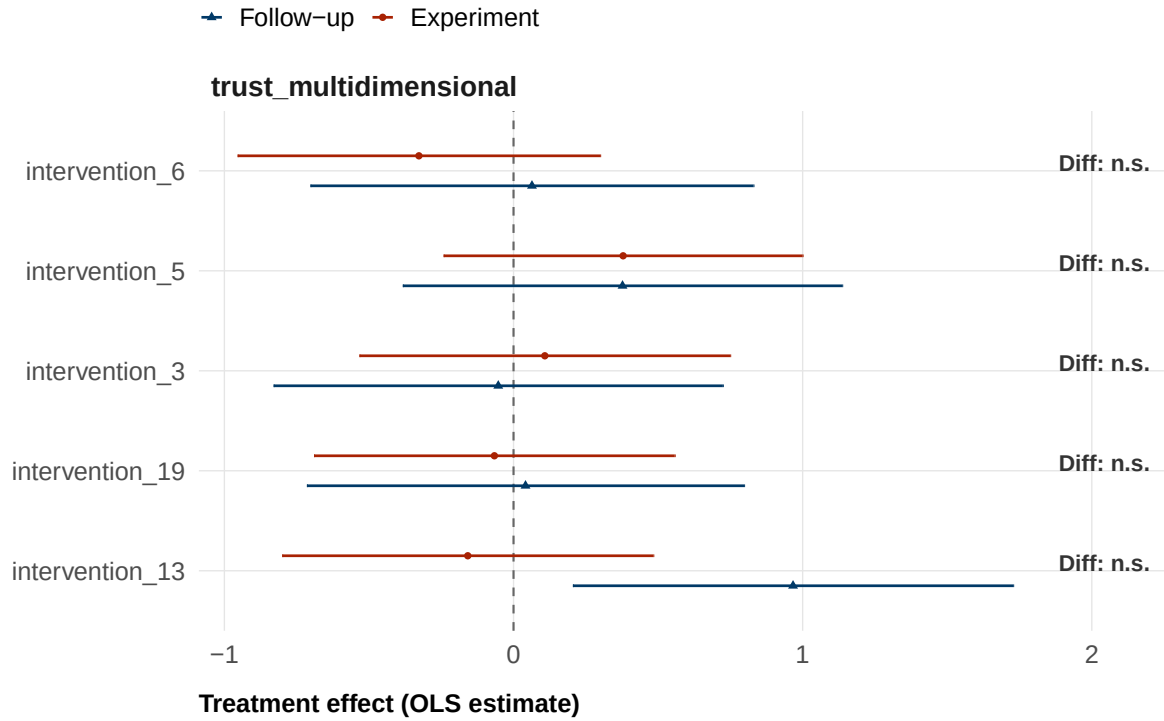



Figure 7: Persistence of treatment effects approximately one week after the experiment, for illustrative outcome variables. Dots represent estimated treatment effects, whiskers depict 95% CIs without correction for multiple comparison. Asterisks next to estimates indicate significance after BH adjustment: * $p_{adj} < .05$; ** $p_{adj} < .01$; *** $p_{adj} < .001$. The ‘Diff’ column indicates whether the difference between experiment and follow-up estimates is statistically significant (same thresholds). Data are simulated, so we should not expect to see any significant effects.

Data and code availability

All code used in this pre-registration and related simulations, as well as all simulated data, is available on [GitHub](#).

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Supplemental Materials

Selection of interventions

The selection process consisted of two steps: first, independent reviews, second, plenary discussion and final selection.

Independent reviews

Each of the 105 submission was independently reviewed by three randomly assigned reviewers⁷. Reviewers provided ratings on four dimensions:

- 1. Theoretical grounding** Is the intervention based on sound and convincing theory?
- 2. Theoretical insight** Would testing this intervention advance theoretical understanding? (Interventions with clear, single mechanisms may offer stronger insight than those combining many mechanisms.)
- 3. Odds of success** How plausible is it that the intervention will work in our study? Helpful questions include: - Has it been tested before? - Was the context comparable (U.S. sample, setting, outcomes)? - How large were the effect sizes?
- 4. Practical relevance** How relevant and scalable is the intervention in the real world? (E.g., can it be implemented easily? Is it feasible at scale?)

All ratings were given on a 0–10 scale (0 = “Not at all / Very weak”; 10 = “Completely / Very strong”). In addition to the numeric ratings, reviewers were encouraged to leave comments to put their ratings into context, and raise points worthwhile discussing during the plenary session.

Based on these independent reviews, we ranked the interventions. For this ranking, we first calculated reviewer specific weighted scores. Each reviewer assigned a weight to each of the four evaluation criteria. As shown in Figure 8, the subjective weights differed considerably between reviewers, but odds of success was considered most important, and theoretical insight least important, on average.

⁷10 reviewers rated 24 interventions, 3 reviewers rated 25 interventions

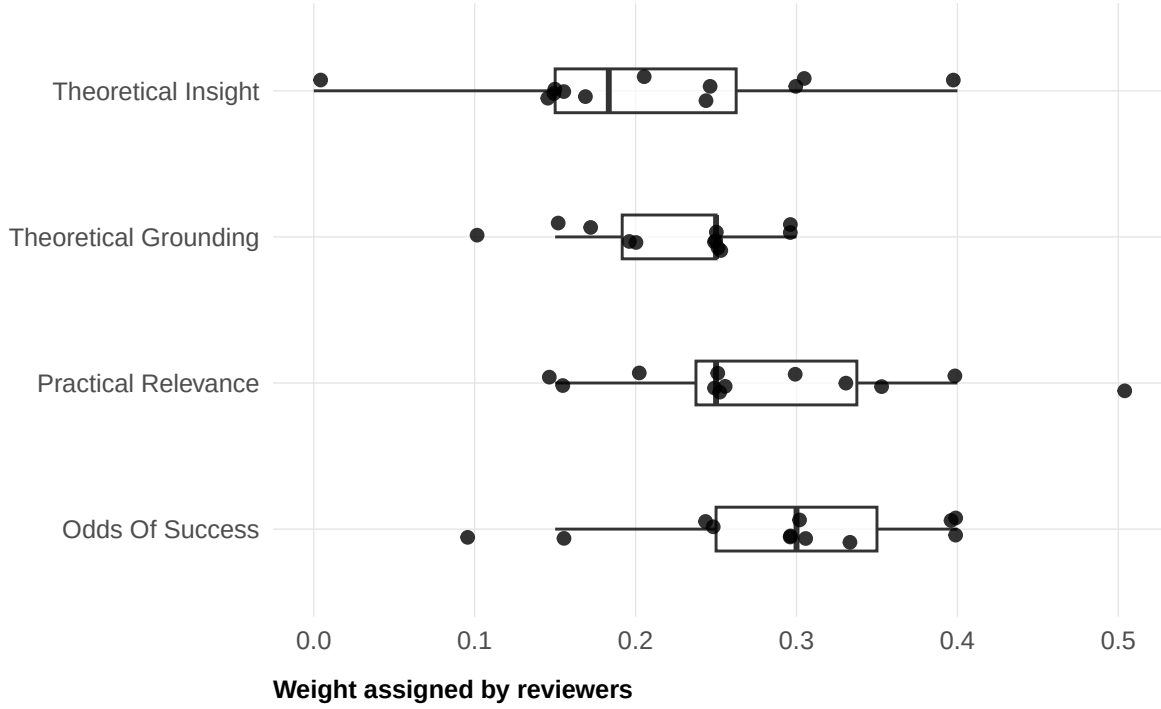


Figure 8: Distributions of weights assigned by reviewers. Each dot corresponds to one reviewer's weight for the respective rating dimension.

For each reviewer, we computed a weighted average score for each intervention they reviewed, based on their own specific weights. Let $w_{r,c}$ be the weight reviewer r gives to criterion c , and $x_{r,i,c}$ be their rating for intervention i on that criterion. Then the review score is:

$$\text{score}_{r,i} = \frac{\sum_{c \in \text{rating_vars}} w_{r,c} \cdot x_{r,i,c}}{\sum_{c \in \text{rating_vars}} w_{r,c}}$$

Since the weights all add up to 1, this simplifies to:

$$\text{score}_{r,i} = \sum_{c \in \text{rating_vars}} w_{r,c} \cdot x_{r,i,c}.$$

We then standardized the reviewers' scores. Reviewers tend to use the rating scales differently: some use the whole scale, some use a narrow band; some generally assign higher scores, some generally lower scores. We therefore standardize the reviewer's scores by z-scoring them. Let: $\text{score}_{r,i}$ be the weighted score reviewer r assigned to intervention i , μ_r be the mean of reviewer r 's scores across all interventions they rated, σ_r be the standard deviation of reviewer r 's scores. Then the standardized score is:

$$\text{zscore}_{r,i} = \frac{\text{score}_{r,i} - \mu_r}{\sigma_r}.$$

This normalization placed all reviews on a common scale in units of (reviewer specific) standard deviation (with mean 0 and standard deviation 1).

Plenary meeting and final selection

We used the standardized aggregate scores from the independent reviews as the baseline for our final selection: by default, the top 20 ranking interventions were selected for testing. However, we held a plenary meeting with the entire research team and advisory board to discuss qualitative reviewer comments and make a final selection. During this meeting we agreed to remove interventions that were not ready to implement and interventions that used fictional characters. To avoid redundancy, we further agreed on merging similar interventions that offered complementary elements. In this case, all teams involved in the merge were contacted to submit a revised intervention together, and all were offered co-authorship. When similar interventions did not show potential to complement each other, we selected the one with the highest rank and removed the others. We also removed two interventions on which major concerns regarding their effectiveness were raised during the discussion, and in qualitative reviewer comments. As a rule, interventions that were removed were replaced by the next-highest ranked intervention. There was only one exception to this rule: an LLM-based discussion intervention. We decided to include this intervention because we had very few dynamic interventions in our selection, and we agreed that the intervention was one of the most promising LLM-based interventions.

Power simulation

For our simulations, we only generate data with a single outcome variable, namely our main outcome trust in climate scientists. To make our simulations more realistic, we used pilot data collected by authors from one of the interventions. We used the control condition from this pilot data ($N = 76$).

In our simulations, we estimated statistical power for different combinations of sample- and effect size parameters. We refer to sample size as the number of participants per experimental condition, with equal sample size assumed for all 20 treatment conditions, and twice as many for the control condition. For example, if sample size $n = 500$, this means $20 \times 500 + 2 \times 500 = 11,000$ participants in total. For each of sample size x effect size combinations, we ran 1,000 simulations. Each simulation generated a data set, which reflected the planned megastudy design with 20 interventions and a shared control group. For more realistic distributions of our outcome variable trust in climate scientists, scores were generated by re-sampling with replacement from the empirical distribution observed in pilot data. No additional error term

was added, as the empirical distribution is assumed to adequately represent the outcome variance in the main study. Treatment effects were simulated by adding a standardized effect size (Cohen’s d) to the intervention conditions, but not to the control condition. For simplicity, all interventions were assumed to have the same effect (i.e., a single common effect size across all 20 interventions).

We translated the standardized effect sizes into point differences on the original 0 to 100 trust scale, by multiplying the standardized effect with the standard deviation of the simulated sample. Because standard deviations could vary slightly between simulations, the translated effect sizes could, too⁸. Outcomes were constrained to the original 0–100 scale. This way, our simulation realistically mimicked potential floor/ceiling effects.

Each simulated dataset was analyzed using a linear regression model predicting the outcome from each experimental condition relative to the control group, yielding 20 different treatment effect estimates.

Statistical significance was evaluated at $\alpha = .05$. Analogous to our analytical procedure, we adjusted p-values for multiple testing via the Benjamini–Hochberg false discovery rate (FDR) procedure. However, for comparison, we also report uncorrected p-values for the power simulations.

Our definition of statistical power differs slightly from the standard definition of power, which is the proportion of simulations in which a *single* effect reaches significance (for a given combination of sample- and effect size). For this megastudy, we defined statistical power as the expected proportion of true intervention effects detected as statistically significant — i.e. the average number of significant intervention effects per simulated dataset, divided by the total number of interventions ($N = 20$). This reflects the study’s ability to identify effective interventions rather than the probability of detecting any single effect.

Simulated sample characteristics

We will provide a descriptive overview of sample characteristics and missing values, similar to Table 4.

Table 4: Summary table for demographic and outcome variables in simulated sample.

Characteristic	N = 22,000
age	46.28 (16.89)
gender	
<i>Male</i>	7,347 (33%)
<i>Female</i>	7,261 (33%)
<i>Other</i>	7,392 (34%)

⁸These changes are minimal in high-powered samples. The expected value of the standard deviations is the standard deviation of the pilot data, $sd = 14.83$.

(continued)

Characteristic	N = 22,000
race	
<i>White / Caucasian</i>	4,369 (20%)
<i>Black / African American</i>	4,450 (20%)
<i>Hispanic / Latino</i>	4,378 (20%)
<i>Asian / Asian American</i>	4,424 (20%)
<i>Other</i>	4,379 (20%)
education	
<i>Less than high school</i>	3,655 (17%)
<i>High school diploma / GED</i>	3,522 (16%)
<i>Some college or Associate's degree</i>	3,672 (17%)
<i>Bachelor's degree</i>	3,775 (17%)
<i>Master's degree / Professional degree</i>	3,663 (17%)
<i>Doctorate degree / Ph.D.</i>	3,713 (17%)
income	
<i>Less than \$30,000</i>	4,388 (20%)
<i>\$30,000 to \$55,999</i>	4,302 (20%)
<i>\$56,000 to \$99,999</i>	4,457 (20%)
<i>\$100,000 to \$167,999</i>	4,425 (20%)
<i>\$168,000 or more</i>	4,428 (20%)
social_class	
<i>Lower class</i>	5,606 (25%)
<i>Working class</i>	5,494 (25%)
<i>Middle class</i>	5,454 (25%)
<i>Upper class</i>	5,446 (25%)
urban_rural	
<i>A large city</i>	5,516 (25%)
<i>A suburb near a large city</i>	5,470 (25%)
<i>A small city or town</i>	5,446 (25%)
<i>A rural area</i>	5,568 (25%)
trust_multidimensional	50.05 (7.83)
<i>Missing</i>	2,382 (11%)
trust_post	50.34 (27.39)
<i>Missing</i>	2,392 (11%)
distrust_post	50.06 (27.56)
<i>Missing</i>	2,422 (11%)
funding_perceptions	50.54 (27.57)
<i>Missing</i>	2,631 (12%)
policy_role_mean	49.92 (13.75)
<i>Missing</i>	2,574 (12%)
inst_trust_mean	49.99 (12.23)
<i>Missing</i>	2,582 (12%)
donation_ams	5.04 (2.88)
<i>Missing</i>	2,803 (13%)
belief_post	49.96 (27.65)
<i>Missing</i>	2,599 (12%)

(continued)

Characteristic	N = 22,000
concern__mean	50.15 (15.81)
<i>Missing</i>	2,581 (12%)
policy__general	50.20 (27.49)
<i>Missing</i>	2,571 (12%)
policy__specific__mean	50.10 (10.49)
<i>Missing</i>	2,757 (13%)
behavior__mean	50.05 (11.70)
<i>Missing</i>	2,787 (13%)
¹ Mean (SD); n (%)	

Statistical Analyses

Primary outcome

Table 5 provides details on the model estimates for the treatment effects on our primary outcome, trust in climate scientists.

Table 5: Treatment effects on trust in climate scientists (multidimensional scale). Interventions are ordered by effect size (largest to smallest). The estimate column reports OLS coefficients with HC2 robust standard errors; significance stars are based on BH-adjusted p-values across all 20 intervention-vs-control comparisons. Both unadjusted and BH-adjusted p-values are reported. 95% confidence intervals are not adjusted for multiple comparisons. In this preregistration, data are simulated at random, thus we should not expect to see any significant effects.

Trust in climate scientists (multidimensional)

	Estimate ^I	SE	95% CI low	95% CI high	p	p (adj.)
intervention_15	0.573	0.316	-0.045	1.192	0.069	0.769
intervention_17	0.567	0.320	-0.061	1.195	0.077	0.769
intervention_5	0.392	0.317	-0.229	1.014	0.216	0.935
intervention_10	0.319	0.319	-0.306	0.944	0.317	0.935
intervention_2	0.316	0.325	-0.322	0.953	0.332	0.935
intervention_9	0.273	0.323	-0.360	0.907	0.397	0.935
intervention_12	0.251	0.328	-0.393	0.895	0.444	0.935
intervention_1	0.115	0.326	-0.523	0.754	0.723	0.935
intervention_3	0.109	0.327	-0.531	0.750	0.739	0.935
intervention_4	0.030	0.319	-0.596	0.655	0.926	0.975
intervention_7	-0.006	0.324	-0.641	0.630	0.986	0.986
intervention_20	-0.063	0.316	-0.683	0.556	0.842	0.935
intervention_19	-0.064	0.318	-0.687	0.560	0.841	0.935
intervention_8	-0.068	0.327	-0.708	0.572	0.835	0.935
intervention_16	-0.086	0.330	-0.732	0.561	0.795	0.935
intervention_11	-0.141	0.321	-0.769	0.487	0.660	0.935
intervention_13	-0.162	0.327	-0.803	0.480	0.621	0.935
intervention_14	-0.164	0.317	-0.785	0.456	0.604	0.935
intervention_6	-0.330	0.320	-0.957	0.297	0.302	0.935
intervention_18	-0.358	0.330	-1.004	0.289	0.278	0.935

^I* p_adj < .05; ** p_adj < .01; *** p_adj < .001 (BH-adjusted). HC2 robust standard errors.

IPW comparison

Figure 9 shows estimated treatment effects using inverse probability weighting (IPW) vs. un-weighted models.

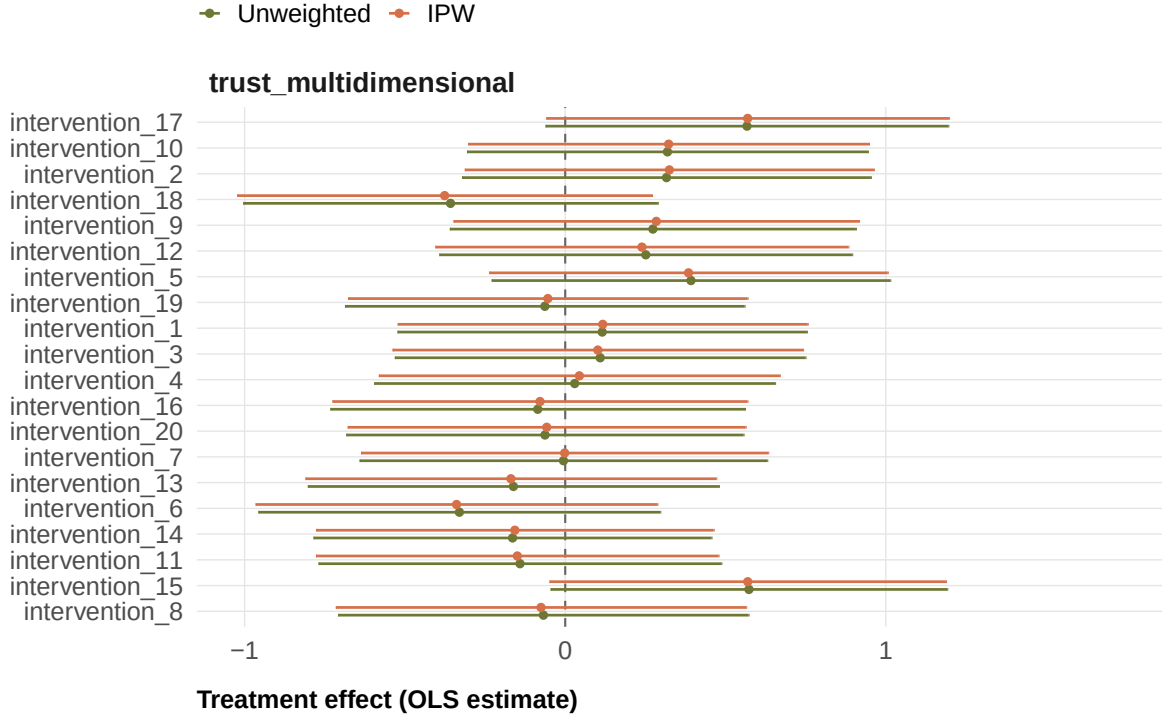


Figure 9: Comparison of estimated treatment effects using inverse probability weighting (IPW) vs. unweighted models, for two illustrative outcome variables. Dots represent estimated treatment effects, whiskers depict 95% CIs without correction for multiple comparison. Asterisks indicate significance after BH adjustment: * $p_{adj} < .05$; ** $p_{adj} < .01$; *** $p_{adj} < .001$. Data are simulated with random attrition, so IPW weights are approximately uniform and differences between the two methods reflect sampling noise only — we should not expect meaningful differences nor significant effects.

Moderators

In this section, we will report estimates of all moderator effects on the primary outcome variable, multidimensional trust in climate scientists. For demonstration, this pre-registration only includes sample tables for one categorical moderator variable, gender, and one continuous moderator variable, age.

Gender

Table 6 shows the moderator effect of gender on the primary outcome variable, multidimensional trust in climate scientists.

Table 6: Moderation of treatment effects by gender. Each cell reports the interaction estimate (BH-adjusted significance stars), unadjusted p-value, and BH-adjusted p-value. Interaction estimates represent the difference in treatment effect for a given gender category relative to the reference category (male). Rows are grouped by intervention, with moderator levels as sub-rows. BH adjustment is applied within each outcome separately across 40 comparisons (20 interventions $\times$ 2 gender categories): * $p_{adj} < .05$; ** $p_{adj} < .01$; *** $p_{adj} < .001$.

Moderation of treatment effects by gender
Estimate* (unadjusted p-value, BH-adjusted p-value)

	Trust (multidimensional)
intervention_1	
Female	-0.39 (p = 0.624, p_{adj} = 0.892)
Other	-0.45 (p = 0.580, p_{adj} = 0.892)
intervention_10	
Female	0.28 (p = 0.724, p_{adj} = 0.921)
Other	0.12 (p = 0.878, p_{adj} = 0.945)
intervention_11	
Female	-1.16 (p = 0.142, p_{adj} = 0.649)
Other	-0.92 (p = 0.243, p_{adj} = 0.649)
intervention_12	
Female	-0.97 (p = 0.239, p_{adj} = 0.649)
Other	-0.45 (p = 0.585, p_{adj} = 0.892)
intervention_13	
Female	-0.09 (p = 0.911, p_{adj} = 0.945)
Other	-0.08 (p = 0.925, p_{adj} = 0.945)
intervention_14	
Female	-1.15 (p = 0.141, p_{adj} = 0.649)
Other	-1.88 (p = 0.015, p_{adj} = 0.586)
intervention_15	
Female	0.23 (p = 0.763, p_{adj} = 0.921)
Other	0.77 (p = 0.326, p_{adj} = 0.814)
intervention_16	
Female	-1.32 (p = 0.108, p_{adj} = 0.649)
Other	-0.17 (p = 0.826, p_{adj} = 0.944)
intervention_17	
Female	-1.16 (p = 0.136, p_{adj} = 0.649)
Other	-1.40 (p = 0.076, p_{adj} = 0.649)
intervention_18	
Female	-1.04 (p = 0.206, p_{adj} = 0.649)
Other	-0.51 (p = 0.530, p_{adj} = 0.892)
intervention_19	
Female	-0.06 (p = 0.941, p_{adj} = 0.945)

Other	-0.93 (p = 0.236, p_adj = 0.649)
intervention_2	
Female	0.49 (p = 0.544, p_adj = 0.892)
Other	0.05 (p = 0.945, p_adj = 0.945)
intervention_20	
Female	-0.26 (p = 0.742, p_adj = 0.921)
Other	-0.61 (p = 0.433, p_adj = 0.866)
intervention_3	
Female	-0.46 (p = 0.563, p_adj = 0.892)
Other	-0.67 (p = 0.417, p_adj = 0.866)
intervention_4	
Female	-1.68 (p = 0.029, p_adj = 0.586)
Other	-0.96 (p = 0.226, p_adj = 0.649)
intervention_5	
Female	0.61 (p = 0.429, p_adj = 0.866)
Other	0.39 (p = 0.614, p_adj = 0.892)
intervention_6	
Female	-0.47 (p = 0.564, p_adj = 0.892)
Other	-1.03 (p = 0.190, p_adj = 0.649)
intervention_7	
Female	-1.05 (p = 0.194, p_adj = 0.649)
Other	-0.35 (p = 0.655, p_adj = 0.903)
intervention_8	
Female	0.33 (p = 0.682, p_adj = 0.910)
Other	0.65 (p = 0.427, p_adj = 0.866)
intervention_9	
Female	-0.22 (p = 0.783, p_adj = 0.921)
Other	-1.01 (p = 0.208, p_adj = 0.649)

* p_adj < .05; ** p_adj < .01; *** p_adj < .001 (BH-adjusted). Stars based on adjusted p-values.

Age

Table 7 shows the moderator effect of gender on the primary outcome variable, multidimensional trust in climate scientists.

Table 7: Moderation of treatment effects by age. Each cell reports the interaction estimate (BH-adjusted significance stars), unadjusted p-value, and BH-adjusted p-value. Interaction estimates represent the difference in treatment effect for a given gender category relative to the reference category (male). Rows are grouped by intervention, with moderator levels as sub-rows. BH adjustment is applied within each outcome separately across 40 comparisons (20 interventions $\times$ 2 gender categories): * p_adj < .05; ** p_adj < .01; *** p_adj < .001.

Moderation of treatment effects by age
Estimate* (unadjusted p-value, BH-adjusted p-value)

	Trust (multidimensional)
intervention_1	
age	0.01 (p = 0.561, p_adj = 0.994)
intervention_10	
age	0.02 (p = 0.248, p_adj = 0.994)
intervention_11	
age	-0.01 (p = 0.468, p_adj = 0.994)
intervention_12	
age	-0.01 (p = 0.706, p_adj = 0.994)
intervention_13	
age	0.01 (p = 0.444, p_adj = 0.994)
intervention_14	
age	0.00 (p = 0.817, p_adj = 0.994)
intervention_15	
age	-0.01 (p = 0.791, p_adj = 0.994)
intervention_16	
age	-0.02 (p = 0.246, p_adj = 0.994)
intervention_17	
age	-0.01 (p = 0.617, p_adj = 0.994)
intervention_18	
age	0.02 (p = 0.319, p_adj = 0.994)
intervention_19	
age	0.01 (p = 0.670, p_adj = 0.994)
intervention_2	
age	0.01 (p = 0.628, p_adj = 0.994)
intervention_20	
age	0.00 (p = 0.894, p_adj = 0.994)
intervention_3	
age	-0.00 (p = 0.895, p_adj = 0.994)
intervention_4	
age	0.00 (p = 0.975, p_adj = 0.997)
intervention_5	
age	-0.04 (p = 0.040, p_adj = 0.791)
intervention_6	
age	-0.03 (p = 0.165, p_adj = 0.994)

intervention_7	
age	-0.02 (p = 0.353, p_adj = 0.994)
intervention_8	
age	0.02 (p = 0.298, p_adj = 0.994)
intervention_9	
age	-0.00 (p = 0.997, p_adj = 0.997)

* p_adj < .05; ** p_adj < .01; *** p_adj < .001 (BH-adjusted). Stars based on adjusted p-values.

Secondary outcomes

Treatment effects

Table 8 provides details on model estimates for secondary outcomes.

Table 8: Treatment effects on secondary outcomes. Each cell reports the estimate (with adjusted significance stars), unadjusted p-value, and BH-adjusted p-value. The intervention is applied within each outcome separately across all comparisons. Data are simulated at random, thus we should not expect significant effects.

	Treatment effects			
	Estimate* (unadjusted p-value, BH-adjusted p-value)			
	Trust (post)	Distrust (post)	Funding perceptions	Policy role
intervention_1	2.65 (p = 0.020, p_adj = 0.409)	0.45 (p = 0.678, p_adj = 0.904)	-0.44 (p = 0.693, p_adj = 0.988)	0.29 (p = 0.618, p_adj = 0.988)
intervention_10	0.11 (p = 0.923, p_adj = 0.950)	1.14 (p = 0.312, p_adj = 0.904)	1.67 (p = 0.138, p_adj = 0.551)	0.54 (p = 0.333, p_adj = 0.988)
intervention_11	1.08 (p = 0.334, p_adj = 0.819)	1.25 (p = 0.268, p_adj = 0.904)	-0.34 (p = 0.764, p_adj = 0.988)	-0.07 (p = 0.907, p_adj = 0.988)
intervention_12	-1.65 (p = 0.140, p_adj = 0.819)	0.76 (p = 0.497, p_adj = 0.904)	0.41 (p = 0.723, p_adj = 0.988)	0.10 (p = 0.856, p_adj = 0.988)
intervention_13	-0.69 (p = 0.532, p_adj = 0.819)	0.56 (p = 0.614, p_adj = 0.904)	1.02 (p = 0.371, p_adj = 0.947)	0.32 (p = 0.561, p_adj = 0.988)
intervention_14	0.14 (p = 0.903, p_adj = 0.950)	0.26 (p = 0.817, p_adj = 0.955)	0.71 (p = 0.535, p_adj = 0.988)	-0.33 (p = 0.558, p_adj = 0.988)
intervention_15	0.80 (p = 0.469, p_adj = 0.819)	3.31* (p = 0.003, p_adj = 0.034)	-0.93 (p = 0.412, p_adj = 0.947)	-0.20 (p = 0.719, p_adj = 0.988)
intervention_16	-0.07 (p = 0.950, p_adj = 0.950)	0.96 (p = 0.407, p_adj = 0.904)	-0.05 (p = 0.965, p_adj = 0.988)	0.07 (p = 0.904, p_adj = 0.988)
intervention_17	1.09 (p = 0.330, p_adj = 0.819)	-0.13 (p = 0.907, p_adj = 0.955)	0.39 (p = 0.727, p_adj = 0.988)	0.53 (p = 0.344, p_adj = 0.988)
intervention_18	0.20 (p = 0.861, p_adj = 0.950)	1.06 (p = 0.355, p_adj = 0.904)	3.87* (p = 0.001, p_adj = 0.013)	-0.83 (p = 0.143, p_adj = 0.988)
intervention_19	1.01 (p = 0.365, p_adj = 0.819)	1.27 (p = 0.260, p_adj = 0.904)	-0.11 (p = 0.927, p_adj = 0.988)	-0.73 (p = 0.198, p_adj = 0.988)
intervention_2	-0.97 (p = 0.377, p_adj = 0.819)	3.68* (p = 0.001, p_adj = 0.020)	1.43 (p = 0.209, p_adj = 0.696)	-0.27 (p = 0.629, p_adj = 0.988)
intervention_20	0.77 (p = 0.487, p_adj = 0.819)	0.47 (p = 0.676, p_adj = 0.904)	-1.79 (p = 0.109, p_adj = 0.547)	-0.10 (p = 0.854, p_adj = 0.988)
intervention_3	0.99 (p = 0.380, p_adj = 0.819)	1.74 (p = 0.120, p_adj = 0.797)	0.38 (p = 0.735, p_adj = 0.988)	0.35 (p = 0.540, p_adj = 0.988)
intervention_4	-1.16 (p = 0.300, p_adj = 0.819)	-0.74 (p = 0.515, p_adj = 0.904)	1.87 (p = 0.101, p_adj = 0.547)	0.29 (p = 0.608, p_adj = 0.988)
intervention_5	0.74 (p = 0.502, p_adj = 0.819)	0.71 (p = 0.524, p_adj = 0.904)	1.87 (p = 0.102, p_adj = 0.547)	-0.19 (p = 0.736, p_adj = 0.988)
intervention_6	0.48 (p = 0.670, p_adj = 0.916)	-0.01 (p = 0.994, p_adj = 0.994)	0.29 (p = 0.800, p_adj = 0.988)	0.03 (p = 0.960, p_adj = 0.988)
intervention_7	-0.41 (p = 0.709, p_adj = 0.916)	0.25 (p = 0.820, p_adj = 0.955)	-0.02 (p = 0.988, p_adj = 0.988)	-0.34 (p = 0.551, p_adj = 0.988)
intervention_8	1.19 (p = 0.293, p_adj = 0.819)	-0.14 (p = 0.898, p_adj = 0.955)	-0.91 (p = 0.426, p_adj = 0.947)	-0.25 (p = 0.659, p_adj = 0.988)
intervention_9	-0.38 (p = 0.732, p_adj = 0.916)	0.65 (p = 0.561, p_adj = 0.904)	0.17 (p = 0.877, p_adj = 0.988)	-0.67 (p = 0.241, p_adj = 0.988)

* p_adj < .05; ** p_adj < .01; *** p_adj < .001 (BH-adjusted). HC2 robust standard errors.

[†]For binary outcomes, estimates are log-odds / average marginal effect in percentage points (pp). P-values are based on log-odds.

Moderators

In this pre-registration, we will only report the moderator effects of gender and age to provide examples.

Gender

Figure 10 and Table 9 show the moderator effect of gender on the secondary outcomes.

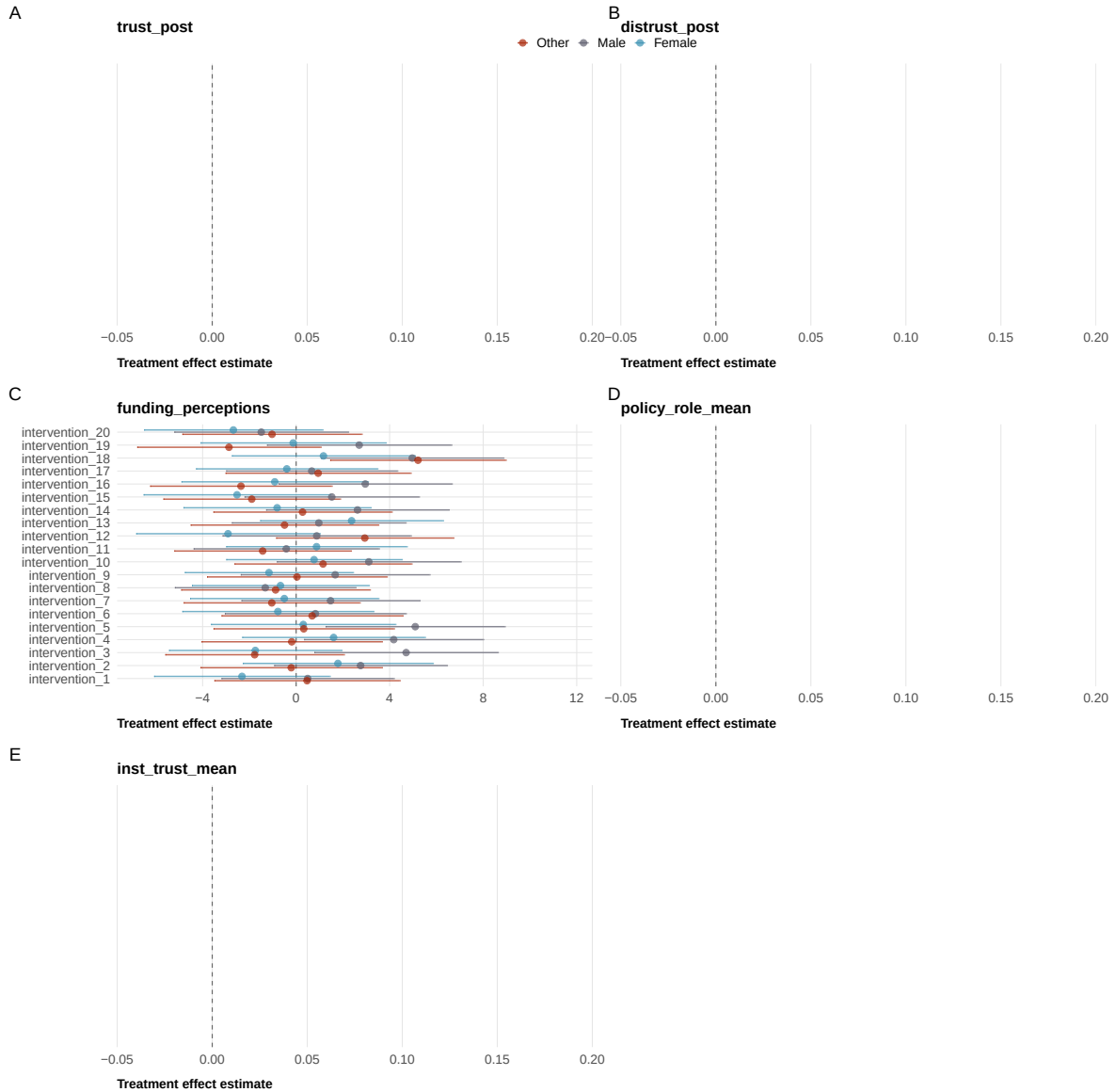


Figure 10: Predicted treatment effects on secondary outcomes by gender. Dots represent predicted effects within each gender category, whiskers depict 95% CIs without correction for multiple comparisons. Significance stars indicate whether the treatment effect in a given category differs significantly from zero, based on BH-adjusted p-values across 60 comparisons (20 interventions $\times$ 3 gender categories) within each outcome. Data are simulated at random, thus we should not expect to see any significant effects.

Table 9: Moderation of treatment effects by gender. Each cell reports the interaction estimate (BH-adjusted significance stars), unadjusted p-value, and BH-adjusted p-value. Interaction estimates represent the difference in treatment effect for a given gender category relative to the reference category (male). Rows are grouped by intervention, with moderator levels as sub-rows. BH adjustment is applied within each outcome separately across 40 comparisons (20 interventions $\times$ 2 gender categories): * p_adj < .05; ** p_adj < .01; *** p_adj < .001.

Moderation of treatment effects by gender
Estimate* (unadjusted p-value, BH-adjusted p-value)

	Funding perceptions	Newsletter signup ^f
intervention_1		
Female	-2.81 (p = 0.296, p_adj = 0.753)	0.05 (p = 0.849, p_adj = 0.917)
Other	-0.03 (p = 0.993, p_adj = 0.993)	-0.15 (p = 0.559, p_adj = 0.799)
intervention_10		
Female	-2.34 (p = 0.399, p_adj = 0.761)	-0.42 (p = 0.116, p_adj = 0.502)
Other	-1.96 (p = 0.482, p_adj = 0.807)	-0.48 (p = 0.060, p_adj = 0.404)
intervention_11		
Female	1.29 (p = 0.647, p_adj = 0.881)	-0.19 (p = 0.465, p_adj = 0.760)
Other	-1.01 (p = 0.718, p_adj = 0.884)	-0.47 (p = 0.070, p_adj = 0.404)
intervention_12		
Female	-3.79 (p = 0.186, p_adj = 0.677)	0.04 (p = 0.892, p_adj = 0.939)
Other	2.06 (p = 0.466, p_adj = 0.807)	-0.60 (p = 0.024, p_adj = 0.237)
intervention_13		
Female	1.40 (p = 0.610, p_adj = 0.872)	-0.26 (p = 0.324, p_adj = 0.681)
Other	-1.47 (p = 0.599, p_adj = 0.872)	-0.35 (p = 0.173, p_adj = 0.502)
intervention_14		
Female	-3.44 (p = 0.229, p_adj = 0.703)	0.09 (p = 0.754, p_adj = 0.900)
Other	-2.35 (p = 0.399, p_adj = 0.761)	-0.13 (p = 0.627, p_adj = 0.845)
intervention_15		
Female	-4.05 (p = 0.146, p_adj = 0.677)	-0.17 (p = 0.513, p_adj = 0.760)
Other	-3.42 (p = 0.207, p_adj = 0.689)	-0.47 (p = 0.071, p_adj = 0.404)
intervention_16		
Female	-3.87 (p = 0.163, p_adj = 0.677)	0.32 (p = 0.217, p_adj = 0.544)
Other	-5.32 (p = 0.052, p_adj = 0.522)	-0.17 (p = 0.505, p_adj = 0.760)
intervention_17		
Female	-1.07 (p = 0.695, p_adj = 0.884)	0.08 (p = 0.747, p_adj = 0.900)
Other	0.27 (p = 0.921, p_adj = 0.978)	-0.09 (p = 0.723, p_adj = 0.900)
intervention_18		
Female	-3.80 (p = 0.178, p_adj = 0.677)	0.08 (p = 0.765, p_adj = 0.900)
Other	0.25 (p = 0.929, p_adj = 0.978)	-0.38 (p = 0.145, p_adj = 0.502)
intervention_19		
Female	-2.82 (p = 0.322, p_adj = 0.753)	0.23 (p = 0.373, p_adj = 0.711)

Other	-5.57 (p = 0.050, p_adj = 0.522)	-0.27 (p = 0.281, p_adj = 0.662)
intervention_2		
Female	-0.97 (p = 0.729, p_adj = 0.884)	-0.25 (p = 0.344, p_adj = 0.687)
Other	-2.97 (p = 0.278, p_adj = 0.753)	-0.39 (p = 0.125, p_adj = 0.502)
intervention_20		
Female	-1.19 (p = 0.660, p_adj = 0.881)	0.06 (p = 0.833, p_adj = 0.917)
Other	0.46 (p = 0.866, p_adj = 0.974)	-0.01 (p = 0.979, p_adj = 0.999)
intervention_3		
Female	-6.45 (p = 0.019, p_adj = 0.410)	0.19 (p = 0.477, p_adj = 0.760)
Other	-6.48 (p = 0.020, p_adj = 0.410)	-0.07 (p = 0.796, p_adj = 0.910)
intervention_4		
Female	-2.57 (p = 0.357, p_adj = 0.753)	-0.82 (p = 0.004, p_adj = 0.071)
Other	-4.36 (p = 0.116, p_adj = 0.663)	-0.26 (p = 0.304, p_adj = 0.676)
intervention_5		
Female	-4.79 (p = 0.088, p_adj = 0.586)	0.66 (p = 0.018, p_adj = 0.237)
Other	-4.77 (p = 0.086, p_adj = 0.586)	0.40 (p = 0.149, p_adj = 0.502)
intervention_6		
Female	-1.60 (p = 0.577, p_adj = 0.872)	-0.33 (p = 0.200, p_adj = 0.534)
Other	-0.14 (p = 0.960, p_adj = 0.985)	-0.85* (p = 0.001, p_adj = 0.041)
intervention_7		
Female	-1.98 (p = 0.484, p_adj = 0.807)	0.19 (p = 0.489, p_adj = 0.760)
Other	-2.51 (p = 0.358, p_adj = 0.753)	-0.00 (p = 0.999, p_adj = 0.999)
intervention_8		
Female	0.65 (p = 0.814, p_adj = 0.957)	0.37 (p = 0.163, p_adj = 0.502)
Other	0.44 (p = 0.877, p_adj = 0.974)	0.13 (p = 0.634, p_adj = 0.845)
intervention_9		
Female	-2.83 (p = 0.305, p_adj = 0.753)	-0.19 (p = 0.484, p_adj = 0.760)
Other	-1.64 (p = 0.565, p_adj = 0.872)	-0.35 (p = 0.176, p_adj = 0.502)

* p_adj < .05; ** p_adj < .01; *** p_adj < .001 (BH-adjusted). Stars based on adjusted p-values.

[†]Estimates are odds ratios from logistic regression.

Age

Figure 11 and Table 10 show the moderator effect of age on the secondary outcomes.

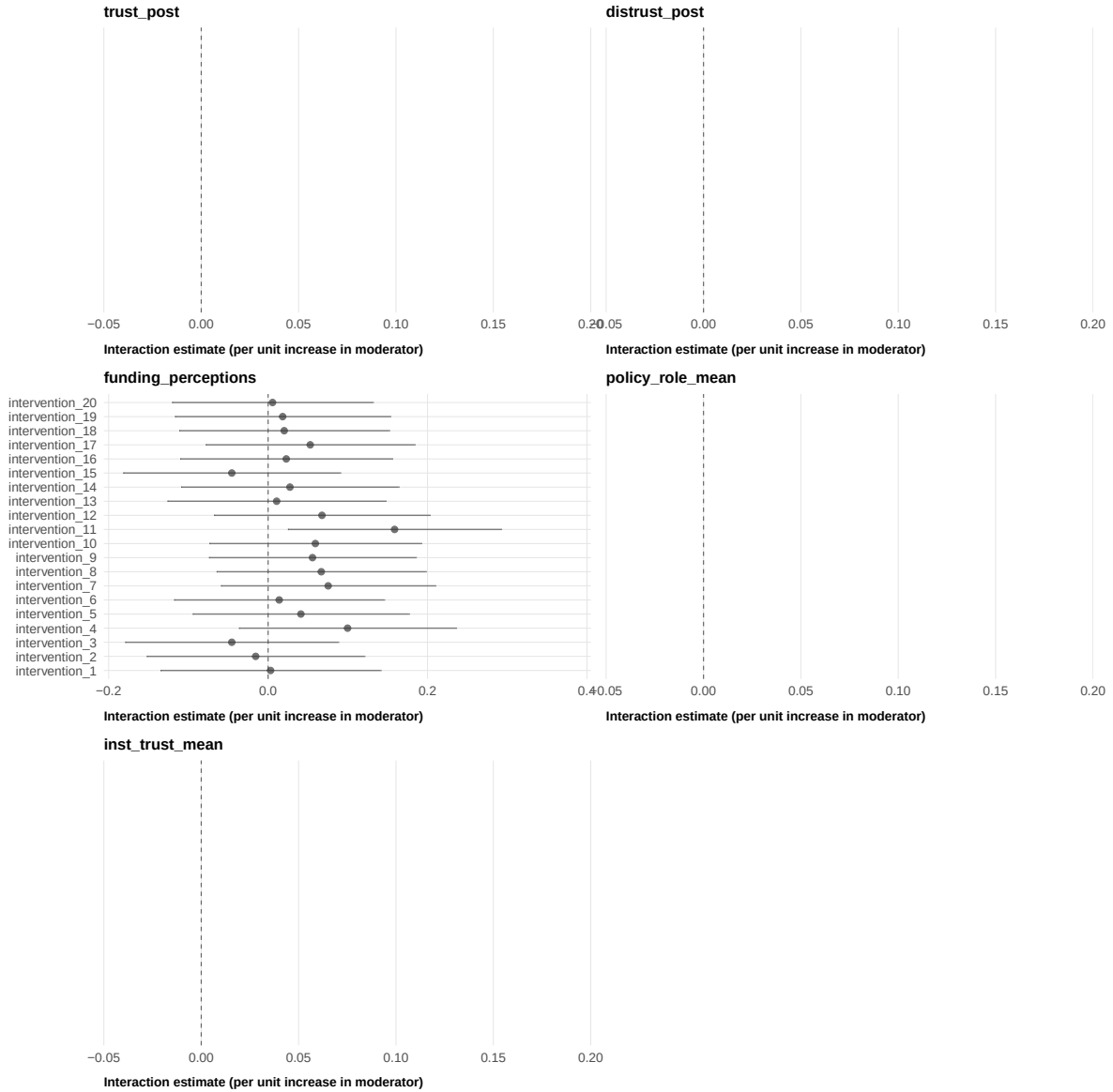


Figure 11: Moderation of treatment effects by age. Each panel shows the interaction estimate per unit increase in age for each outcome. A positive estimate indicates the intervention is more effective for older participants. Dots represent interaction estimates, whiskers depict 95% CIs without correction for multiple comparisons. Asterisks indicate significance after BH adjustment: * $p_{adj} < .05$; ** $p_{adj} < .01$; *** $p_{adj} < .001$. Data are simulated at random, thus we should not expect to see any significant effects.

Table 10: Moderation of treatment effects by age. Each cell reports the interaction estimate (BH-adjusted significance stars), unadjusted p-value, and BH-adjusted p-value. Interaction estimates represent the difference in treatment effect for a given gender category relative to the reference category (male). Rows are grouped by intervention, with moderator levels as sub-rows. BH adjustment is applied within each outcome separately across 40 comparisons (20 interventions $\times$ 2 gender categories):
* p_adj < .05; ** p_adj < .01; *** p_adj < .001.

Moderation of treatment effects by age
Estimate* (unadjusted p-value, BH-adjusted p-value)

	Funding perceptions
intervention_1	
age	0.00 (p = 0.964, p_adj = 0.964)
intervention_10	
age	0.06 (p = 0.382, p_adj = 0.964)
intervention_11	
age	0.16 (p = 0.020, p_adj = 0.401)
intervention_12	
age	0.07 (p = 0.328, p_adj = 0.964)
intervention_13	
age	0.01 (p = 0.878, p_adj = 0.964)
intervention_14	
age	0.03 (p = 0.693, p_adj = 0.964)
intervention_15	
age	-0.05 (p = 0.512, p_adj = 0.964)
intervention_16	
age	0.02 (p = 0.737, p_adj = 0.964)
intervention_17	
age	0.05 (p = 0.429, p_adj = 0.964)
intervention_18	
age	0.02 (p = 0.763, p_adj = 0.964)
intervention_19	
age	0.02 (p = 0.791, p_adj = 0.964)
intervention_2	
age	-0.02 (p = 0.823, p_adj = 0.964)
intervention_20	
age	0.01 (p = 0.931, p_adj = 0.964)
intervention_3	
age	-0.05 (p = 0.504, p_adj = 0.964)

intervention_4	
age	0.10 (p = 0.151, p_adj = 0.964)
intervention_5	
age	0.04 (p = 0.552, p_adj = 0.964)
intervention_6	
age	0.01 (p = 0.836, p_adj = 0.964)
intervention_7	
age	0.08 (p = 0.272, p_adj = 0.964)
intervention_8	
age	0.07 (p = 0.319, p_adj = 0.964)
intervention_9	
age	0.06 (p = 0.400, p_adj = 0.964)

* p_adj < .05; ** p_adj < .01; *** p_adj < .001 (BH-adjusted). Stars based on adjusted p-values.

Persistence

Figure 12 provides a visualization of the treatment effects in the experiment and the follow-up survey, Table 11 provides more details.

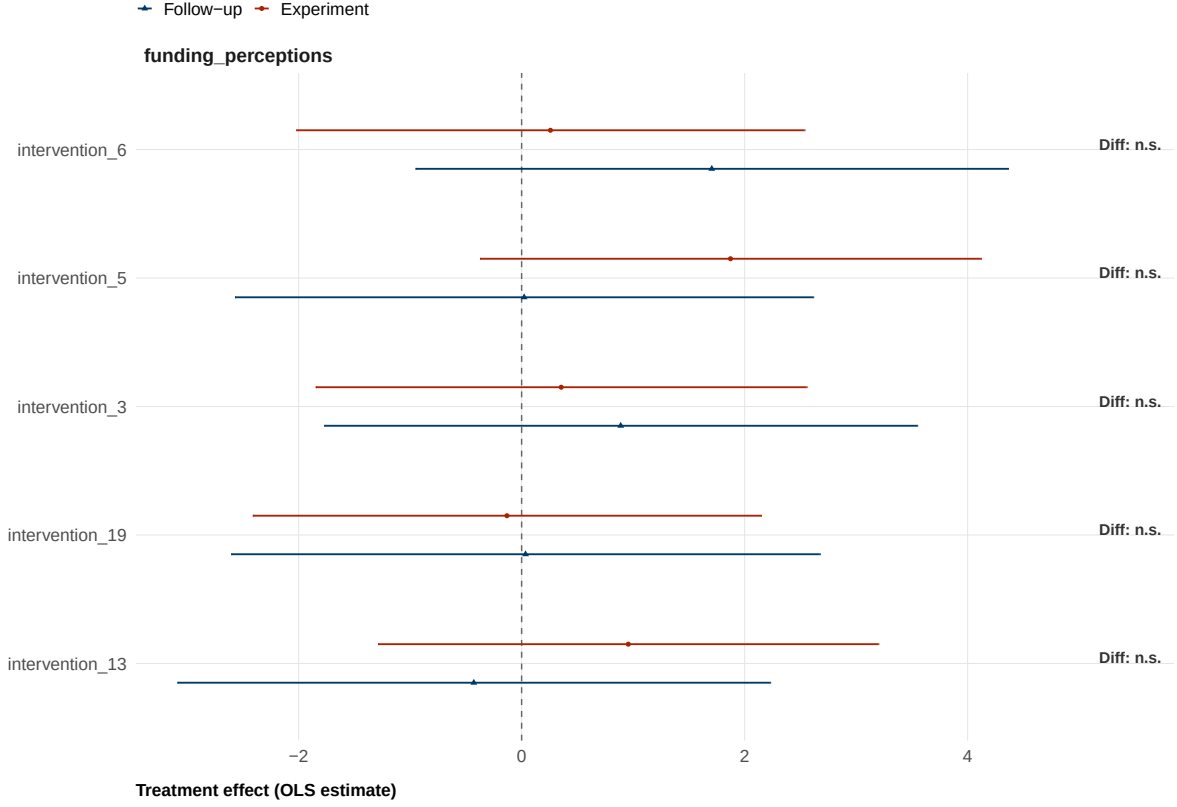


Figure 12: Persistence of treatment effects on secondary outcomes approximately one week after the experiment. Dots represent estimated treatment effects, whiskers depict 95% CIs without correction for multiple comparison. Asterisks next to estimates indicate significance after BH adjustment: * $p_{adj} < .05$; ** $p_{adj} < .01$; *** $p_{adj} < .001$. The ‘Diff’ column indicates whether the difference between experiment and follow-up estimates is statistically significant (same thresholds). Data are simulated, so we should not expect to see any significant effects.

Table 11: Persistence of treatment effects on secondary outcomes. Each cell reports the condition $\times$ time interaction estimate (BH-adjusted significance stars), unadjusted p-value, and BH-adjusted p-value. The interaction term tests whether the treatment effect changed between the experiment and the follow-up survey approximately one week later. A negative estimate indicates the effect diminished at follow-up. BH adjustment is applied within each outcome separately across all 20 intervention-vs-control comparisons. Data are simulated at random, thus we should not expect to see any significant effects.

Persistence of treatment effects — secondary outcomes

Estimate* (unadjusted p-value, BH-adjusted p-value)

	Funding perceptions
intervention_13	
Experiment	0.96 (p = 0.403, p_adj = 0.986)
Follow-up	-0.43 (p = 0.752, p_adj = 0.986)
Interaction	-1.39 (p = 0.441, p_adj = 0.736)
intervention_19	
Experiment	-0.13 (p = 0.910, p_adj = 0.986)
Follow-up	0.04 (p = 0.979, p_adj = 0.986)
Interaction	0.17 (p = 0.927, p_adj = 0.927)
intervention_3	
Experiment	0.35 (p = 0.752, p_adj = 0.986)
Follow-up	0.89 (p = 0.512, p_adj = 0.986)
Interaction	0.53 (p = 0.766, p_adj = 0.927)
intervention_5	
Experiment	1.87 (p = 0.102, p_adj = 0.986)
Follow-up	0.02 (p = 0.986, p_adj = 0.986)
Interaction	-1.85 (p = 0.294, p_adj = 0.736)
intervention_6	
Experiment	0.26 (p = 0.824, p_adj = 0.986)
Follow-up	1.71 (p = 0.208, p_adj = 0.986)
Interaction	1.45 (p = 0.421, p_adj = 0.736)

* p_adj < .05; ** p_adj < .01; *** p_adj < .001 (BH-adjusted). 'Interaction' tests whether the treatment effect changed between experiment and follow-up. 'Experiment' and 'Follow-up' show predicted effects within each wave.

Tertiary outcomes

In the manuscript, we will report additional analyses and regression coefficient tables in the same way as done for the secondary outcomes. For brevity, we have not run these analyses in this preregistration.

Item-level analyses

Several outcome scales comprise multiple items that may respond differently to the interventions. We analyze item-level heterogeneity for five scales: trust dimensions, specific climate policies, individual-level behaviors, institutional trust, and climate change concern. For each scale, we estimate a single interaction model on long-format data, stacking observations across all items. Standard errors are clustered at the participant level. We report two sets of results: predicted treatment effects within each item, and interaction terms testing whether effects differ across items relative to the reference item (first level of each scale).

Formally, for all item-level models:

$$Y_{ij} = \beta_0 + \sum_{k=1}^K \beta_k D_{ik} + \beta_J Item_j + \sum_{k=1}^K \beta_{kJ} (D_{ik} \times Item_j) + \varepsilon_{ij}$$

where Y_{ij} is outcome i on item j , D_{ik} are intervention dummies with control as reference, $Item_j$ are item dummies with one item as reference, β_k is the treatment effect on the reference item, β_{kJ} tests whether the effect differs on item j relative to the reference, and $\beta_k + \beta_{kJ}$ is the predicted effect on item j . We adjust for multiple comparison within each scale. For example, for the predicted treatment effects of the different trust dimensions, we apply multiple comparison correction across all 80 comparisons (20 interventions $\times$ 4 dimensions) testing whether effects differ from zero within each dimension. This adjustment is conservative, as it treats the dimensions as one family of tests. For the interaction effects testing whether effects differ across dimensions relative to the (omitted) baseline dimensions, in this case competence, we apply multiple comparison correction to all 60 interaction terms (20 interventions $\times$ 3 dimensions)⁹, again treating the all dimensions as one family of tests.

```
# trust dimensions
trust_dimensions_results <- run_items_model(
  data      = data,
  items     = trust_dimensions,
  outcome_name = "trust",
  covariates = covariates
)

# specific climate policies
policy_items_results <- run_items_model(
  data      = data,
  items     = policy_specific_items,
  outcome_name = "policy_specific",
  covariates = covariates
)

# individual behaviors
behavior_items_results <- run_items_model(
  data      = data,
  items     = behavior_items,
  outcome_name = "behavior",
  covariates = covariates
)

# institutional trust
```

⁹One dimension is omitted because it is used as the baseline category.

```

inst_trust_items_results <- run_items_model(
  data      = data,
  items     = inst_trust_items,
  outcome_name = "inst_trust",
  covariates = covariates
)

# concern: absolute vs relative
data <- data |>
  mutate(concern_absolute = rowMeans(pick(concern_1, concern_2), na.rm = TRUE))

concern_items_results <- run_items_model(
  data      = data,
  items     = c("concern_absolute", "concern_3"),
  outcome_name = "concern",
  covariates = covariates
)

```

Trust dimensions

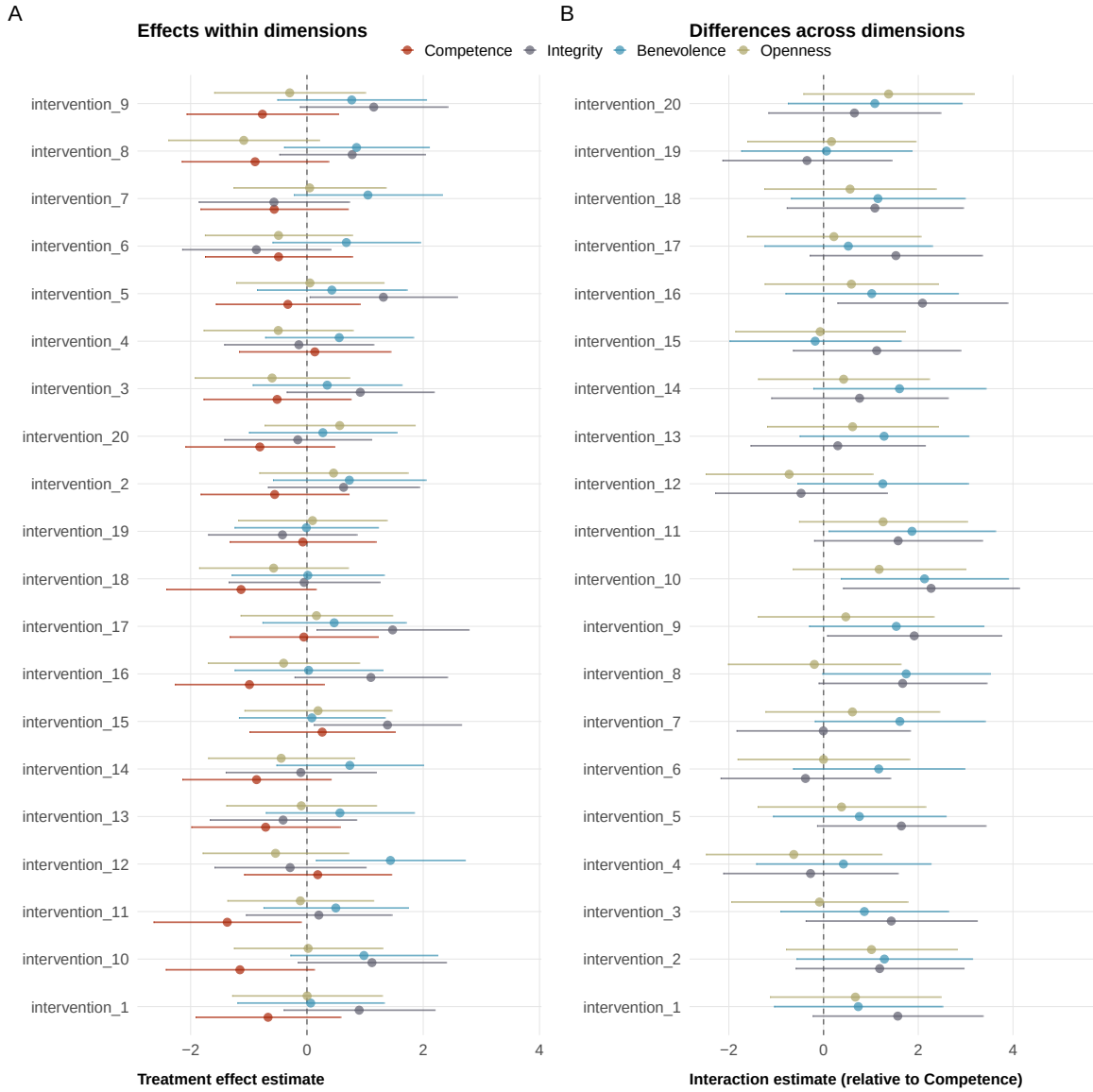


Figure 13: Treatment effects on trust subdimensions. **A** Predicted treatment effects within each dimension, BH-adjusted across 80 comparisons (20 interventions $\times$ 4 dimensions). **B** Interaction terms testing whether effects differ across dimensions relative to competence (reference), BH-adjusted across 60 interaction terms. Whiskers depict 95% CIs. Data are simulated at random.

Specific climate policies

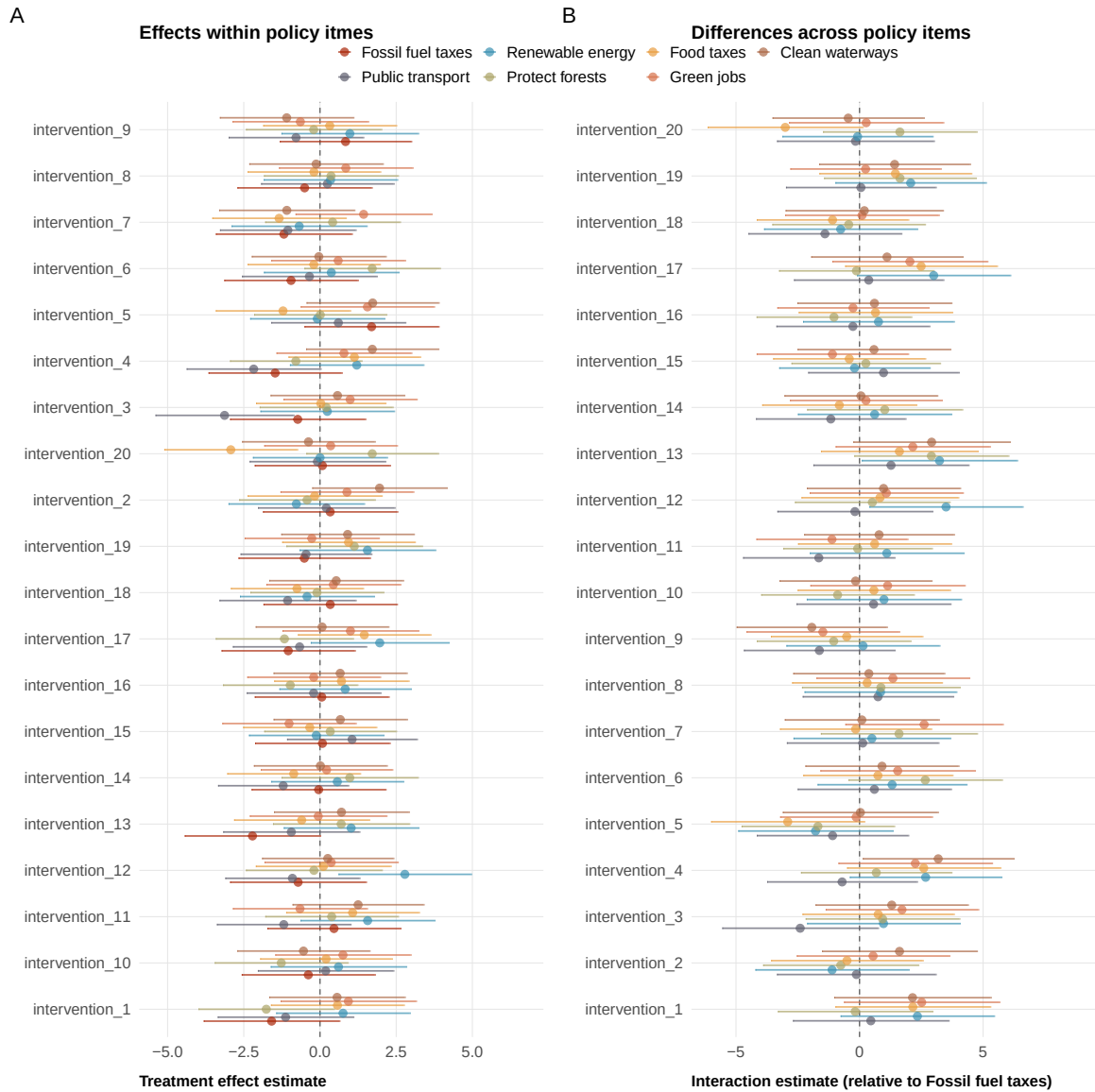


Figure 14: Treatment effects on specific climate policy items. **A** Predicted treatment effects within each policy item, BH-adjusted across 140 comparisons (20 interventions $\times$ 7 items). **B** Interaction terms testing whether effects differ across policy items relative to the reference item (fossil fuel taxes), BH-adjusted across 120 interaction terms. Whiskers depict 95% CIs. Data are simulated at random.

Individual-level behaviors

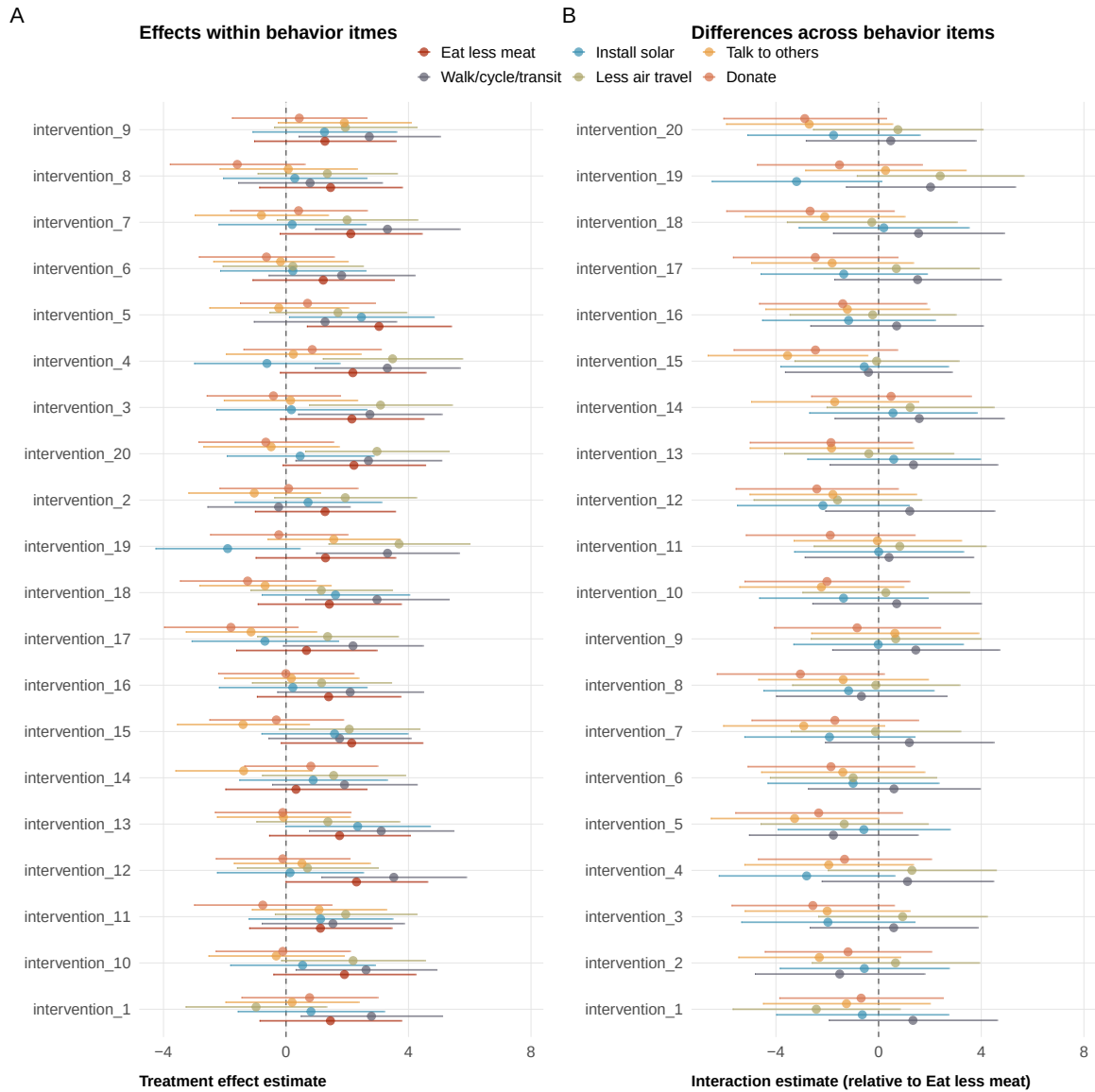


Figure 15: Treatment effects on individual-level climate behavior items. **A** Predicted treatment effects within each behavior item, BH-adjusted across 120 comparisons (20 interventions $\times$ 6 items). **B** Interaction terms testing whether effects differ across behavior items relative to the reference item (eat less meat), BH-adjusted across 100 interaction terms. Whiskers depict 95% CIs. Data are simulated at random.

Institutional trust

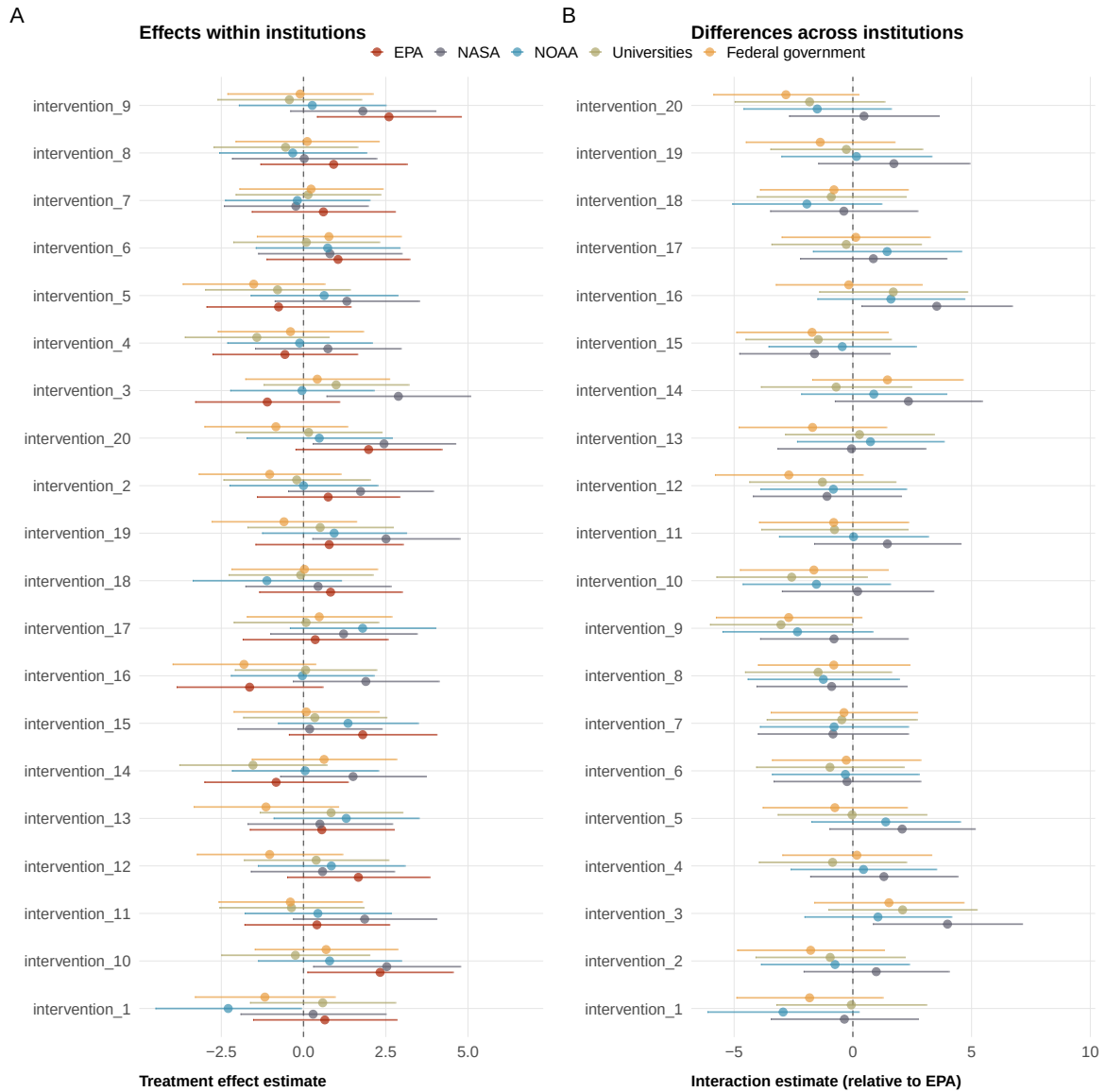


Figure 16: Treatment effects on institutional trust items. **A** Predicted treatment effects within each institution, BH-adjusted across 100 comparisons (20 interventions $\times$ 5 institutions). **B** Interaction terms testing whether effects differ across institutions relative to the reference item (EPA), BH-adjusted across 80 interaction terms. Whiskers depict 95% CIs. Data are simulated at random.

Climate change concern: absolute vs. relative

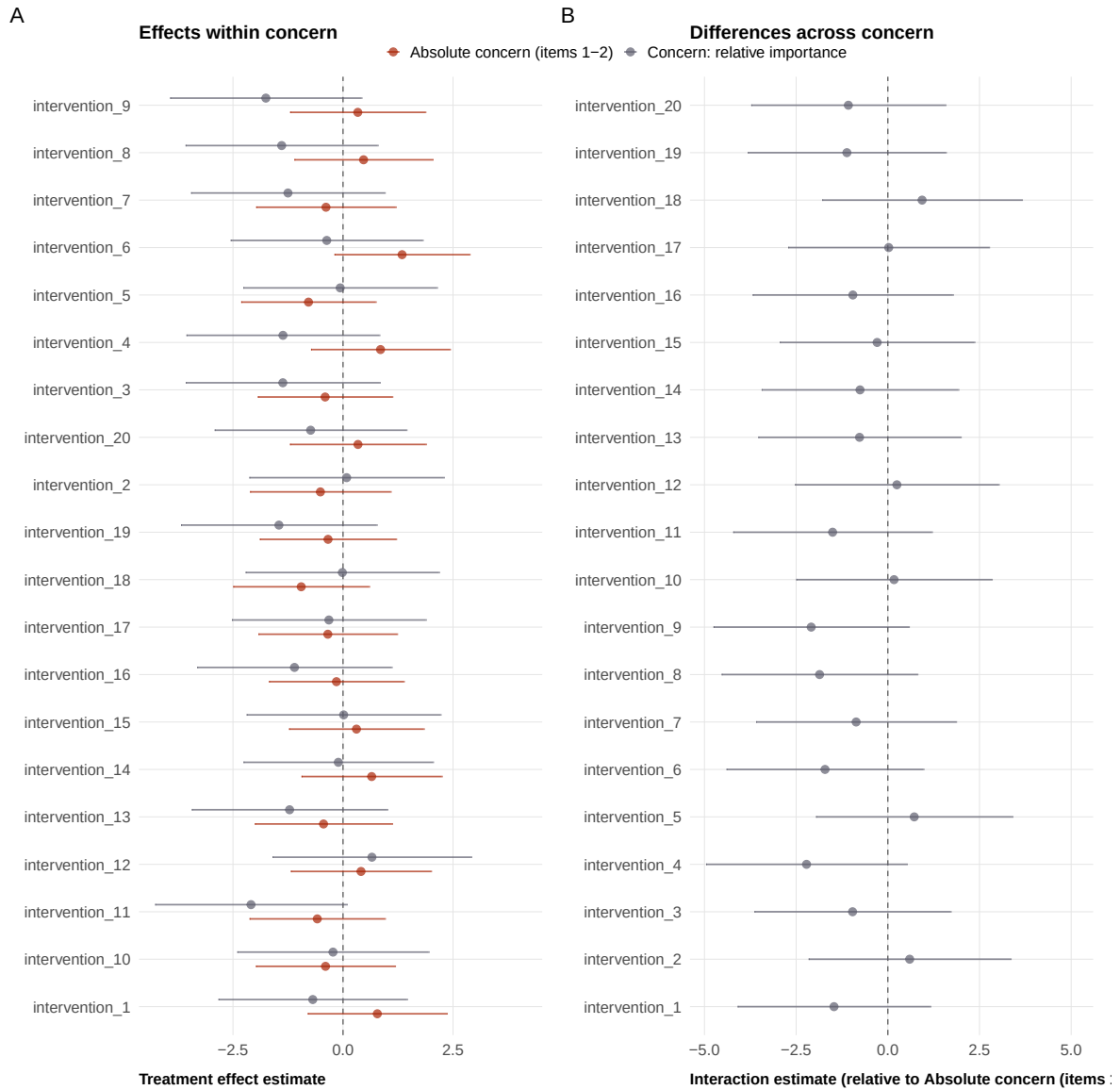


Figure 17: Treatment effects on climate change concern items. **A** Predicted treatment effects by absolute vs. relative concern, BH-adjusted across 40 comparisons (20 interventions $\times$ 2 institutions). **B** Interaction terms testing whether effects differ absolute and relative concern items, BH-adjusted across 20 interaction terms. Whiskers depict 95% CIs. Data are simulated at random.

Scale properties

We report scale properties for all multi-item outcome measures to document the internal consistency of our scales (Table 12). All statistics are computed on the control group only (N = 2,000) to avoid contamination from differential treatment effects on individual items.

We report two complementary indicators of internal consistency. Cronbach’s α is the most widely used reliability measure, reflecting the average inter-item correlation weighted by the number of items. It increases with both the number of items and the average inter-item correlation, which makes it sensitive to scale length. We also report the mean inter-item correlation (mean r), which is independent of scale length.

In addition, we report inter-item correlation matrices for each scale—for trust in Table 13, for institutional trust in Table 14, for concern in Table 15, for policy preferences in Table 16, for behaviors in Table 17. We expect items to be heterogeneous for essentially every scale. The correlation matrices will provide insight on whether this expectation was correct. Low inter-item correlations would justify treating items separately rather than as a composite, as is done in the previous section.

Table 12: Internal consistency of multi-item scales computed on the control group only (N 2,000). Cronbach's α and mean inter-item correlation (mean r) are reported. Data are simulated at random, thus values are not meaningful.

Scale	N items	Cronbach's α	Mean r
Trust: All dimensions	12	0.00	0.00
Trust: Competence	3	-0.04	-0.01
Trust: Integrity	3	0.03	0.01
Trust: Benevolence	3	-0.03	-0.01
Trust: Openness	3	0.08	0.03
Institutional trust	5	0.00	0.00
Policy role	4	0.03	0.01
Concern	3	-0.01	0.00
Specific policies	7	0.05	0.01
Individual behaviors	6	0.01	0.00

Table 13: Inter-item correlations for all trust items across all four subdimensions (control group only). Items are grouped by subdimension.

	Open- ness 3	Open- ness 2	Open- ness 1	Benev- o- lence 3	Benev- o- lence 2	Benev- o- lence 1	In- tegrity 3	In- tegrity 2	In- tegrity 1	Com- pe- tence 3	Com- pe- tence 2
Com- pe- tence 1	-0.012	0.029	0.012	0.005	0.002	0.02	0.035	-0.011	-0.005	-0.028	-0.014
Com- pe- tence 2	-0.037	0.006	0.014	-0.032	-0.02	-0.019	0.003	-0.007	-0.032	0.002	
Com- pe- tence 3	-0.007	-0.024	0.004	-0.003	0.006	-0.011	0.025	0.006	0.022		
In- tegrity 1	0.011	-0.05	-0.025	0.029	0.003	0.029	0.006	0.026			
In- tegrity 2	0.043	-0.001	0.005	-0.007	-0.009	0.035	0.002				
In- tegrity 3	-0.011	-0.009	-0.024	-0.024	-0.012	-0.024					
Benev- o- lence 1	0.009	0.025	-0.006	-0.028	-0.014						
Benev- o- lence 2	0.001	0.008	-0.037	0.017							
Benev- o- lence 3	-0.014	0.036	0.012								
Open- ness 1	0.037	0.023									
Open- ness 2	0.026										

Table 14: Inter-item correlations for institutional trust items (control group only).

	Federal government	Universities	NOAA	NASA
EPA	-0.043	-0.014	0.008	-0.021
NASA	-0.002	0.011	-0.019	
NOAA	0.017	0.01		
Universities	0.048			

Table 15: Inter-item correlations for concern items (control group only). Note item 3 (relative importance) is expected to correlate less strongly with items 1–2 (absolute concern).

	Concern: relative importance	Concern: how serious?
Concern: how concerned?	-0.025	0.011
Concern: how serious?	0.006	

Table 16: Inter-item correlations for specific climate policy items (control group only).

	Clean waterways	Green jobs	Food taxes	Protect forests	Renewable energy	Public transport
Fossil fuel taxes	-0.009	0.011	0.023	-0.007	0.056	0.03
Public transport	0.034	0.023	0.004	-0.005	0.005	
Renewable energy	0.013	0.008	-0.026	0.005		
Protect forests	0.031	-0.002	-0.037			
Food taxes	0.021	-0.009				
Green jobs	-0.007					

Table 17: Inter-item correlations for individual climate behavior items (control group only).

	Donate	Talk to others	Less air travel	Install solar	Walk/cycle/transit
Eat less meat	-0.025	-0.015	0.001	-0.013	0
Walk/cycle/transit	-0.015	0.027	-0.001	-0.017	
Install solar	0.004	-0.005	0.06		
Less air travel	-0.003	0.035			
Talk to others	-0.017				